Fluid Mechanics.

- A substance which can flow is called Fluid.
- Substances having fixed volume or density but shape can change.
- Liquids + Gases

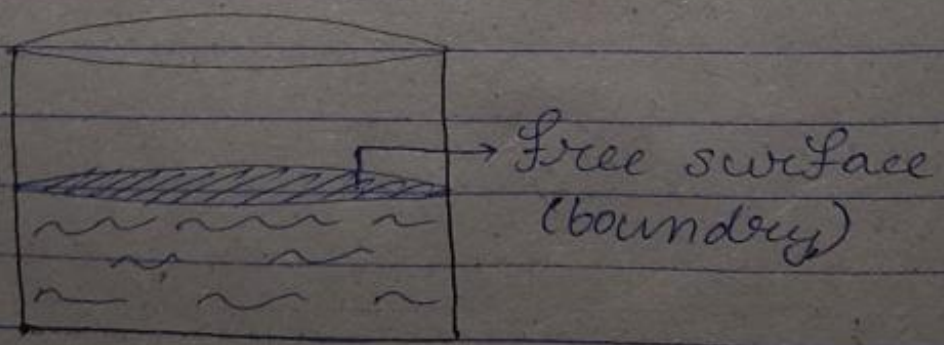
Gases are considered fluids only when they are incompressible.

Assumptions:

(i) Fluid is incompressible, i.e fixed volume and density whatever pressure be applied.

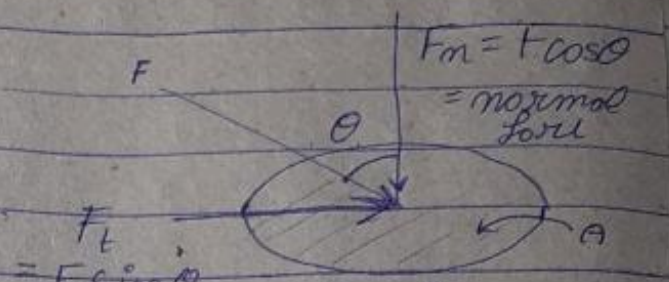
(ii) They are non viscous, ie. no resistive force in tangential direction by liquid layers.

(iii) Fluid will always have a boundary with solids or gases.



① Pressure

$$P = \frac{F_n}{A} = \frac{F\cos\theta}{A}$$



Fluids doesn't sustain sheer stress.

⇒ shear stress $= \frac{F_t}{A} = \frac{F\sin\theta}{A}$



$P = \frac{dF}{dA}$ ⇒ Fluids layers apply force on each other only along common normal.

Unit: $N/m^2 = J/m^3 = Pa$.

⇒ 1 atm. $= 1.013 \times 10^5\ N/m^2$ = 76 cm of Hg
= 760 mm of Hg
= 760 torrs

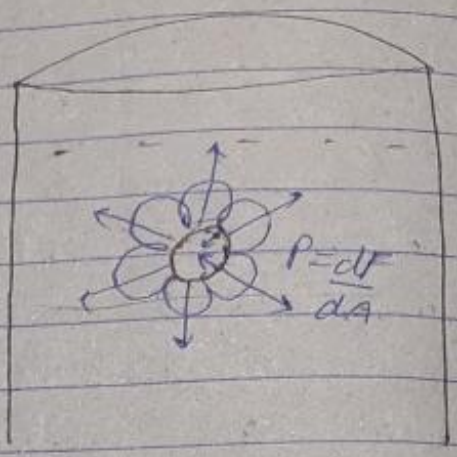
1 bar $= 10^5\ N/m^2$ or Pa

It is a scalar quantity.

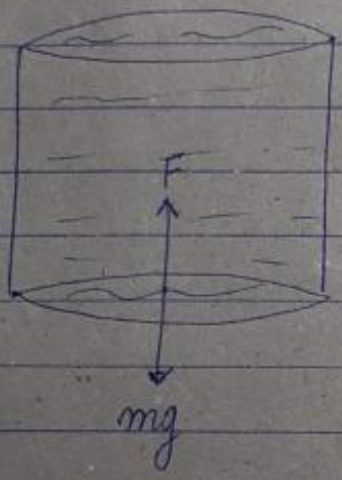
dim: $[ML^{-1}T^{-2}]$

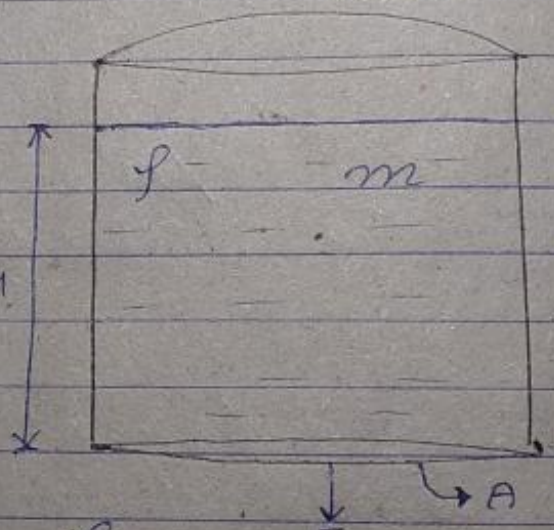
② Fluid Pressure

Pressure applied by molecules on each other due to gravitational pull.

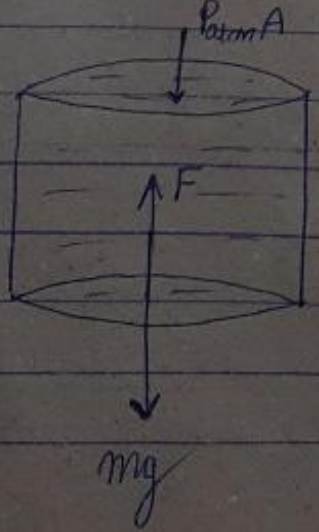


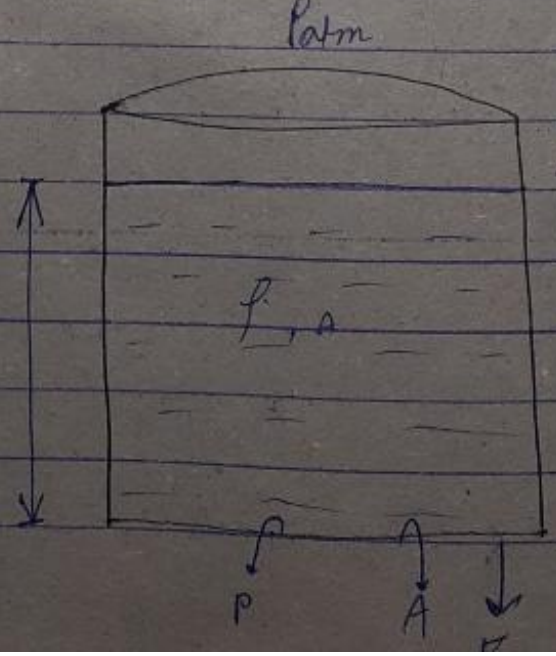
(i) Pressure due to Gravity





$$P = \frac{F}{A} = \frac{mg}{A} = \frac{\rho AHg}{A} = \rho gH$$





$F - mg - P_{atm}A = ma$

$F = m(g+a) + P_{atm}A$

$\therefore P = \frac{F}{A} = \frac{m(g+a)}{A} + P_{atm} = \rho h(g+a) + P_{atm}$

③ Relative density / Specific Gravity.

$R.D = \frac{\text{density of substance}}{\text{density of water at } 4^\circ C}$ (or) reference substn.

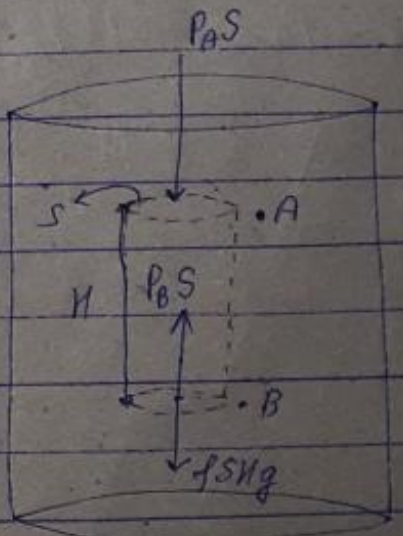
density of H_2O at $4^\circ C = 1\ gm/cm^3$

R.D is unitless or dimentionless.

(eg) $\rho_{alcohol} = 800\ kg/m^3 \Rightarrow RD = \frac{800}{1000} = 0.8$

④ Variation of Pressure with height

(C1) Container is at rest.



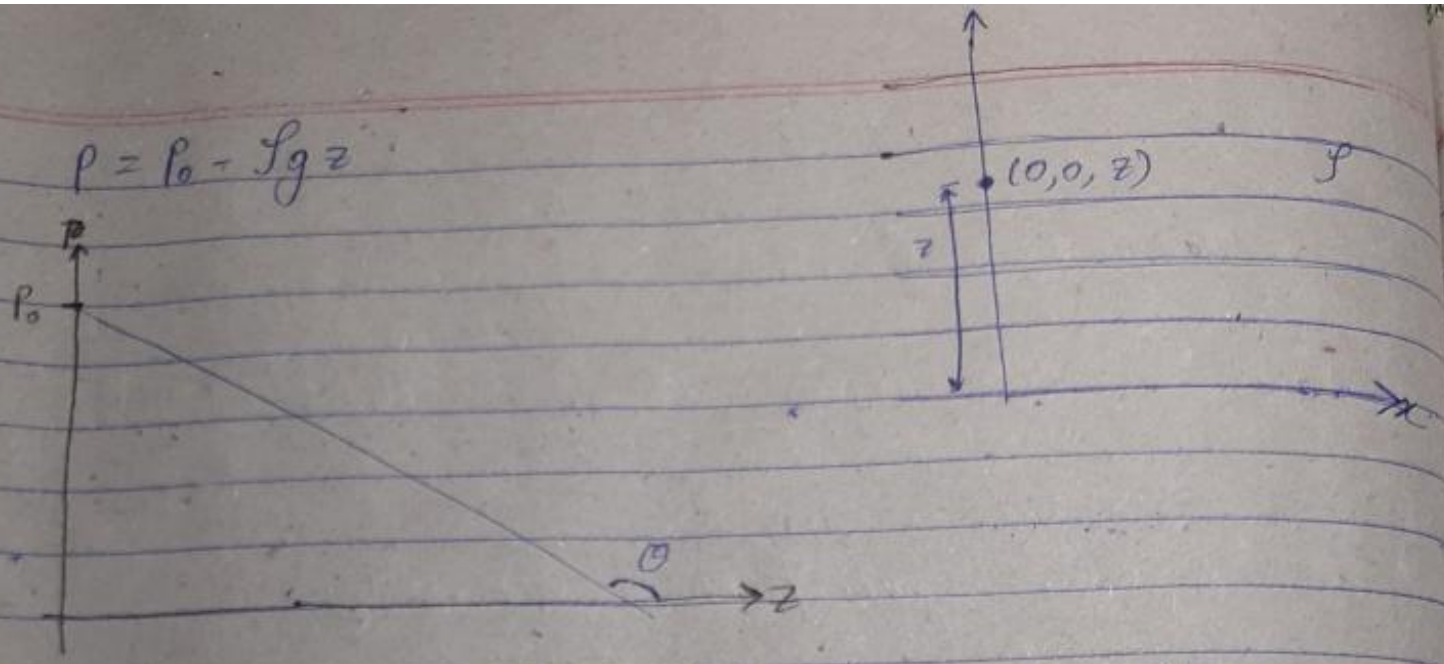
$P_0S + \rho SHg = P_BS$

$\boxed{P_B = P_A + \rho gH}$

$\Delta P = P_B - P_A = \rho gH$

Pressure increase going down and decrease going up.

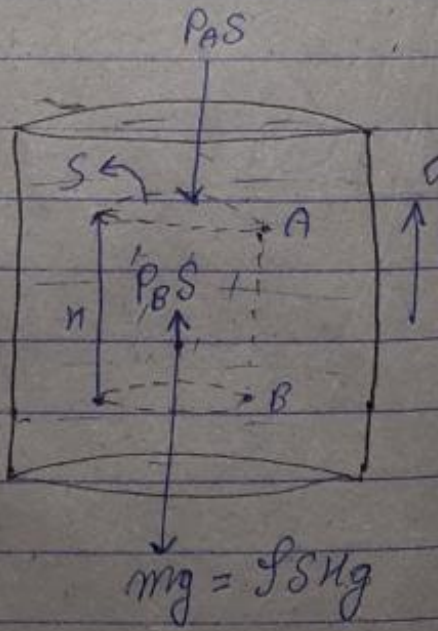
$P = P_0 - \rho g z$



$\tan\theta = m = -\rho g$ = Pressure Gradient.

$\frac{dP}{dz} = -g\rho$

(2) Container is accelerating vertically upwards.



$P_B S - P_A S - \rho S H g = (\rho S H) a$

$P_B - P_A = \rho H (g + a)$

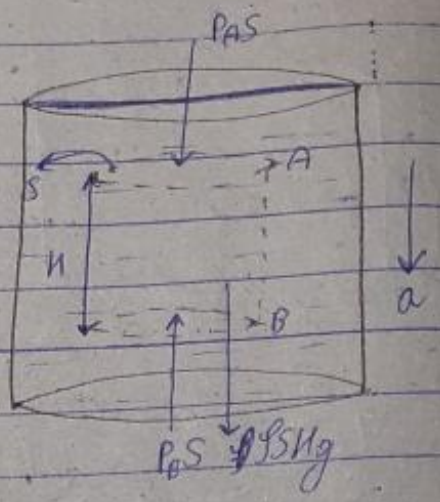
$= \rho H g_{eff}$

(3) Container is accelerating downwards.

$P_A S + \rho S H g - P_B S = \rho S H a$

$P_B - P_A = \rho H (g - a)$

$= \rho H g_{eff}$

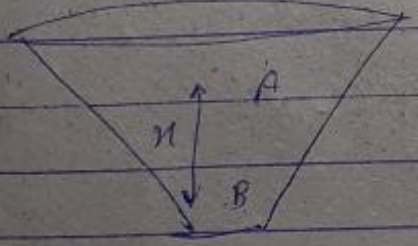


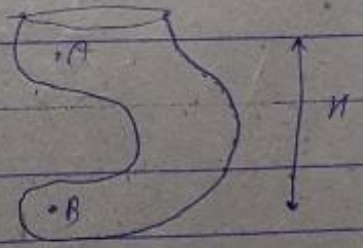
if $g > a \Rightarrow P_B > P_A$

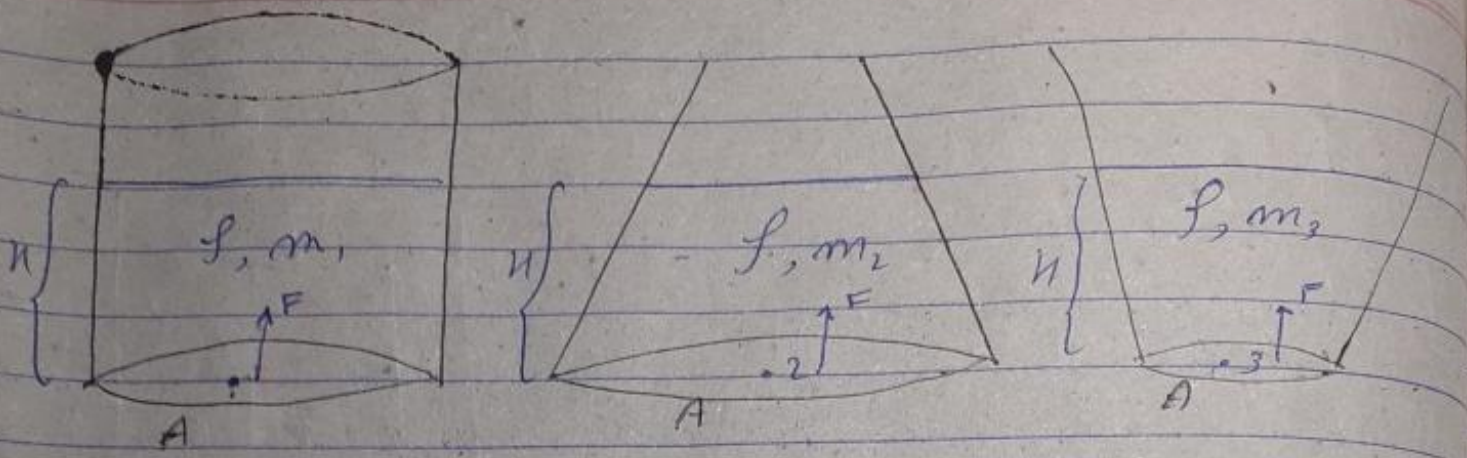
if $g = a \Rightarrow P_B = P_A$

if $g < a \Rightarrow P_A > P_B$

These results are independent of shape of container.





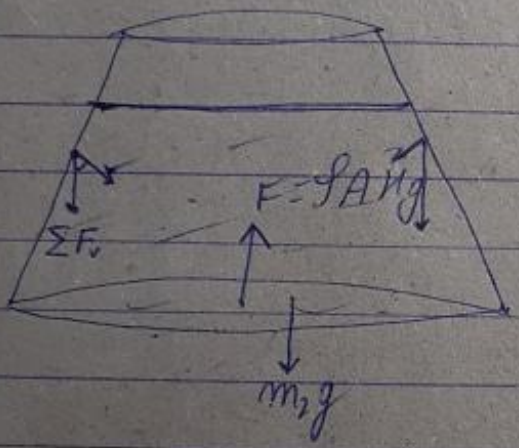


$P_1 = P_2 = P_3$ $\quad F = \rho g H A$

$m_3 > m_1 > m_2$

$m_1 g = \rho A H g$

②

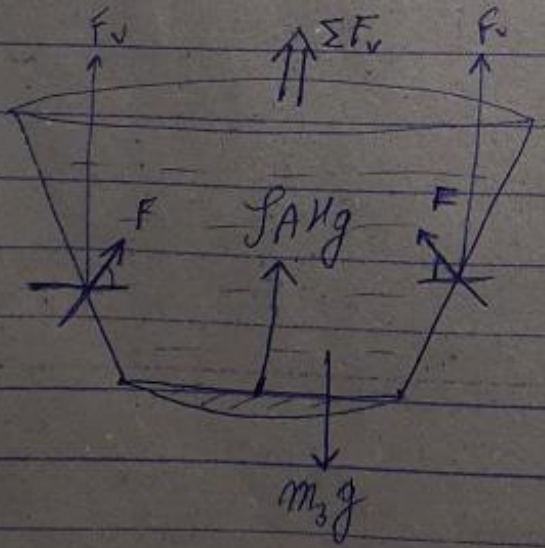


$F = m_2 g + \sum F_v$

$\sum F_v = F - m_2 g$

$\sum F_v = \rho g H A - m_2 g$

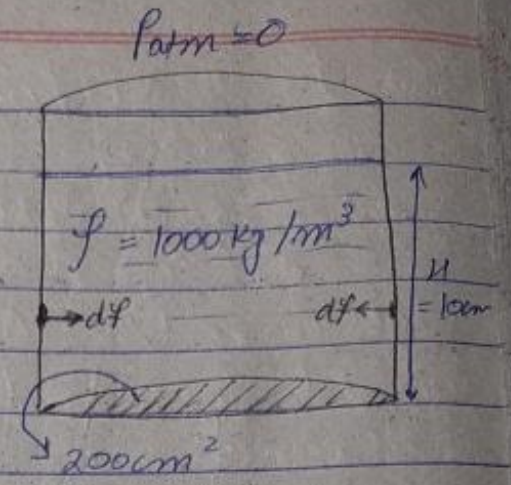
③



$\sum F_v + \rho A H g = m_3 g$

$\sum F_v = m_3 g - \rho A H g$

(8) Find force applied by base of container and vertical wall of container.



$F_{vertical} = 0$

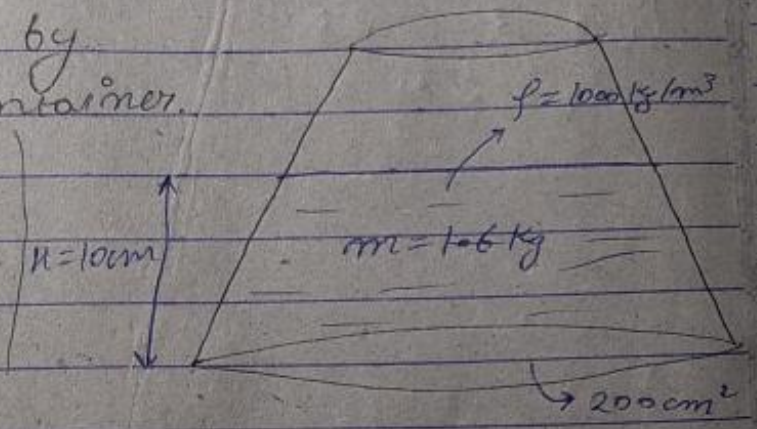
$\rho H g A = mg$

$\Rightarrow 1000 \times 10 \times 0.1 \times 200 \times 10^{-4}$

$= 20\ N$

(8) Find force applied by liquid on base of container and on the ~~top~~ slant sides of container.

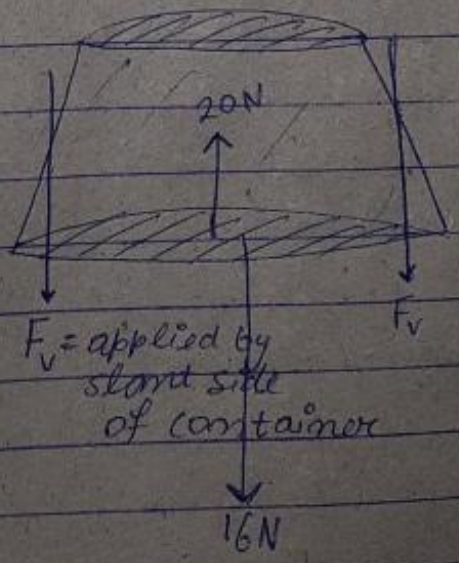
$P_{atm} = 0$.



$F = PA = \rho g H A$

$= 10^3 \times 10 \times 200 \times 10^{-4}$

$= 20\ N$



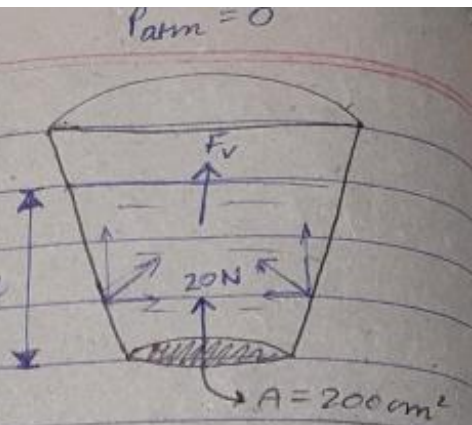
$20 = 16 + F_V$

$F_V = 4\ N$

(Q) $\rho = 1000\ kg/m^3$

$m = 2.5\ kg$

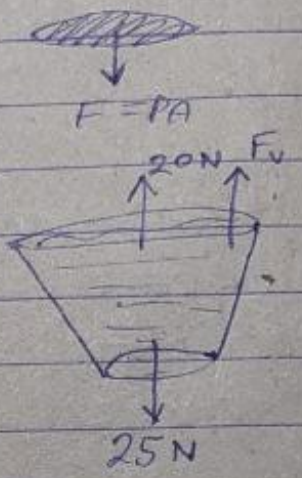
Find force applied by base of container on water and also find force applied by slant height of container on wall.



$F = PA = \rho g H A = 20\ N$

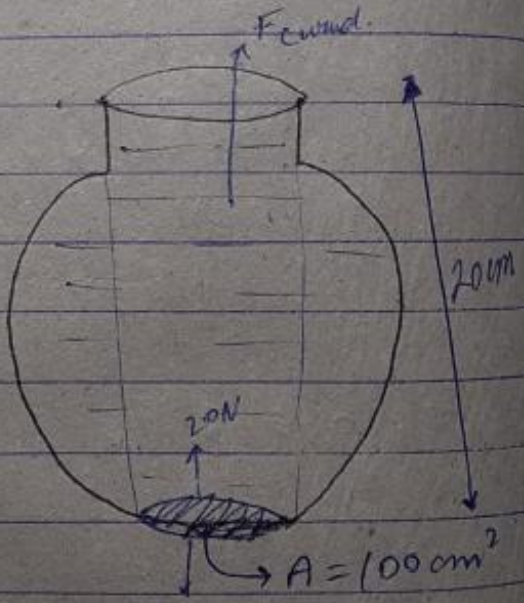
$F_v + 20 = 25$

$F_v = 5\ N$



(Q) $m = 3\ kg$, $\rho = 1000\ kg/m^3$

Find force applied by base of container on water Also find force applied by curved section of container on water



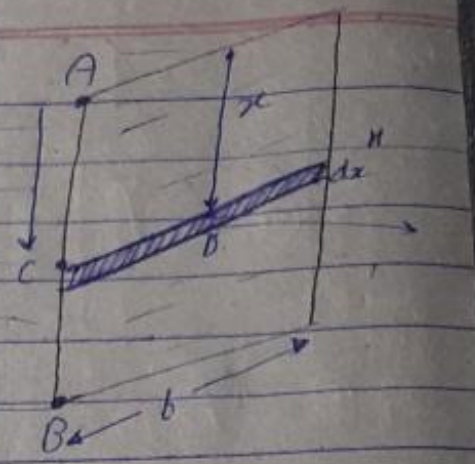
$F = PA = \rho g H A = 20\ N$

$F + 20 = 30$

$F = 10\ N$

P_{atm}

Dam



Q Find Pressure at points A, B, C, D.

at A: $P_A = P_{atm}$

$P_B = P_{atm} + \rho g H$

$P_C = P_{atm} + \rho g x = P_D$

Find force applied by liquid on the wall.

$$dF = P\,dA = (P_{atm} + \rho g x)\, b\, dx$$

$$F = \int_0^F dF = P_{atm}\, b \int_0^H dx + \rho g b \int_0^H x\, dx$$

$$\boxed{F = P_{atm}\, b H + \rho g b \frac{H^2}{2}}$$

Force due to atmosphere on the inner wall of dam.

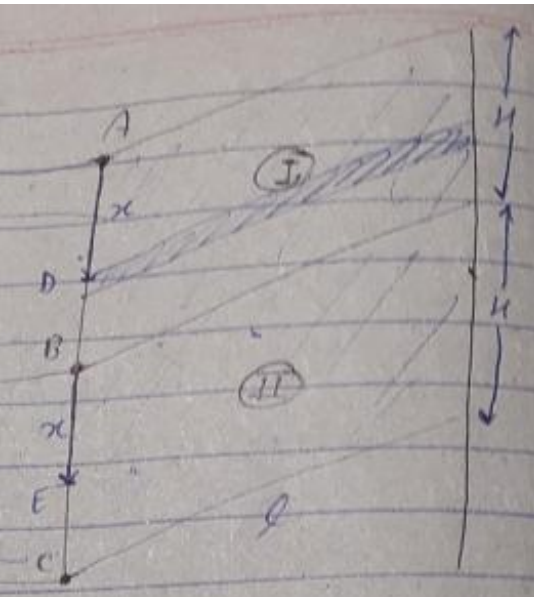
$$F = P_{atm}\, b H$$

$P_{atm} = 0$

(6)

Liq (A) ρ

Liq (B) 2ρ



Find Pressure at Points A, B, C, D, E.

$P_A = 0$, $P_B = \rho g H$, $P_C = \rho g H + 2\rho g H = 3\rho g H$.

$P_D = \rho g x$, $P_E = \rho g H + 2\rho g x$.

Find, ① Force applied by liquid A on face I,

② " " " B on face II

③ " " A " " I $= 0$

④ " " A " " II $= 0$

① $dF = \rho g x l \, dx$

$$\therefore F = \int_0^F dF = \int_0^H \rho g x l \, dx = \rho g l \int_0^H x \, dx$$

$$= \rho g l \frac{H^2}{2}$$

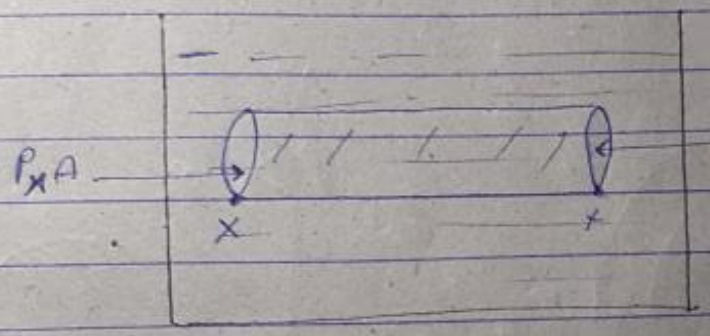
② $dF = \rho g H + 2\rho g x \Rightarrow F = \int_0^H \rho g H + 2\rho g x$

$$\therefore F = \rho g H \int_0^H dx + \rho g \int_0^H x\,dx$$

$$F = \rho g H H + \rho g \frac{H^2}{2} \Rightarrow \frac{3}{2}\rho g H^2 \quad 2\rho g H^2 l$$

Find force on II due to liquid A = $\rho g H$

- In a connected homogeneous liquid which is unaccelerated in horizontal direction, points at same horizontal level will have same pressure.

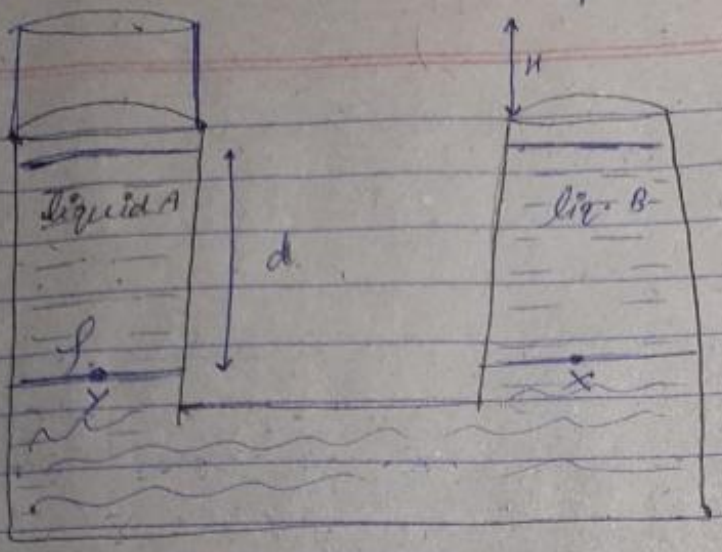


In horizontal direction, $a = 0$

$\therefore P_x A = P_y A$

$P_x = P_y$

(Q)

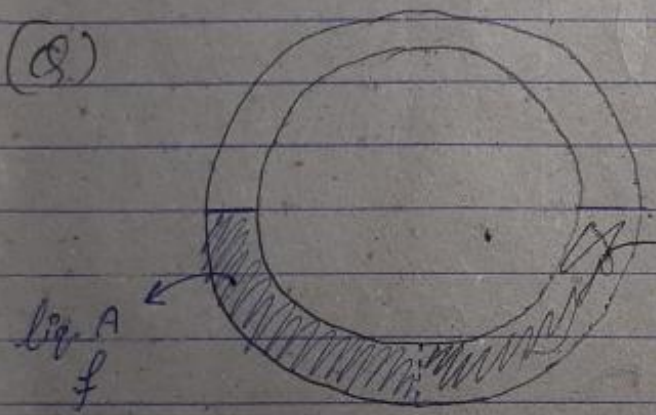


Find ρ_B

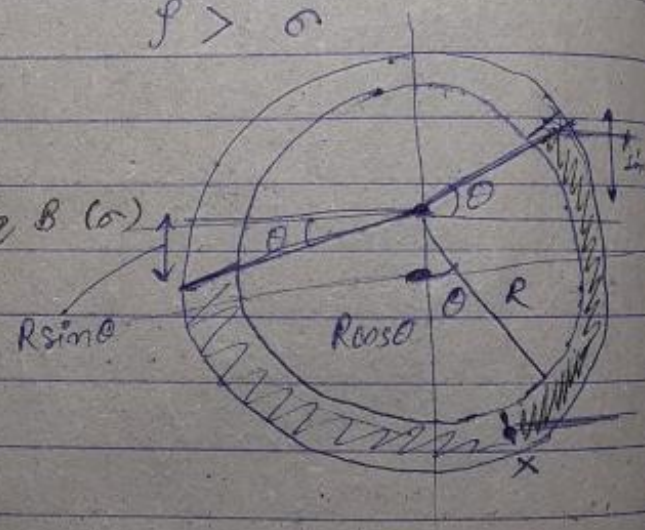
$P_x = P_y \Rightarrow P_{atm} + \rho g(H+d) = P_{atm} + \sigma g d$

$\sigma = \frac{\rho(H+d)}{d}$

(Q)



$\rho > \sigma$



⇒ find θ

$P_x = 0 + \rho g h = \rho g(R\cos\theta - R\sin\theta) = \rho g R(\cos\theta - \sin\theta)$

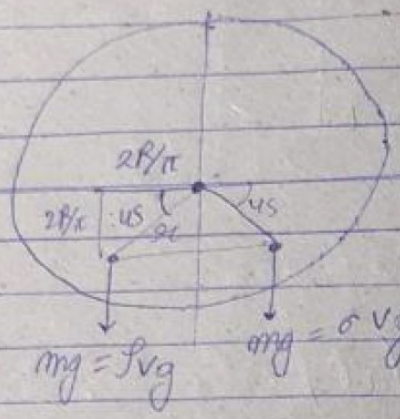
$= \sigma g H = \sigma g R(\cos\theta + \sin\theta)$

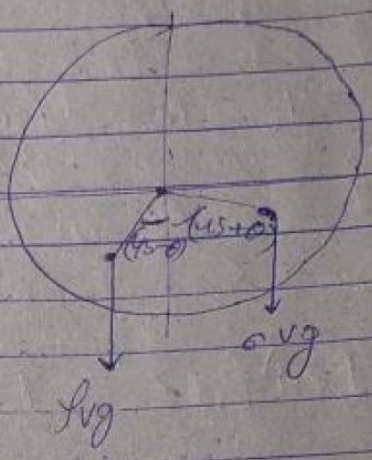
$\Rightarrow \frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta} = \frac{\rho}{\sigma} \Rightarrow \frac{1+\tan\theta}{1-\tan\theta} = \frac{\rho}{\sigma}$

$\sigma + \sigma\tan\theta = \rho - \rho\tan\theta$

$(\sigma + \rho)\tan\theta = (\rho - \sigma)$

$$\tan\theta = \frac{\sigma - \rho}{\sigma + \rho} \quad \frac{\rho - \sigma}{\rho + \sigma} \Rightarrow \theta = \tan^{-1}\left(\frac{\rho - \sigma}{\rho + \sigma}\right)$$

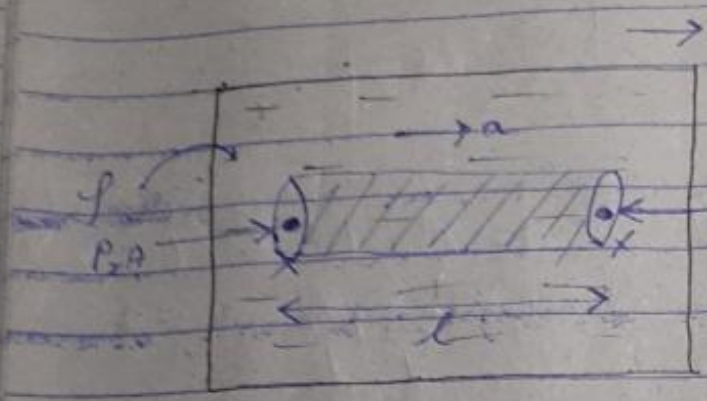




$$\Rightarrow \rho Vg \times \sin(45-\theta) = \sigma Vg \times \sin(45+\theta)$$

$$\frac{\sin(45-\theta)}{\sin(45+\theta)} = \frac{\sigma}{\rho}$$

• Pressure difference in accelerating liquid.

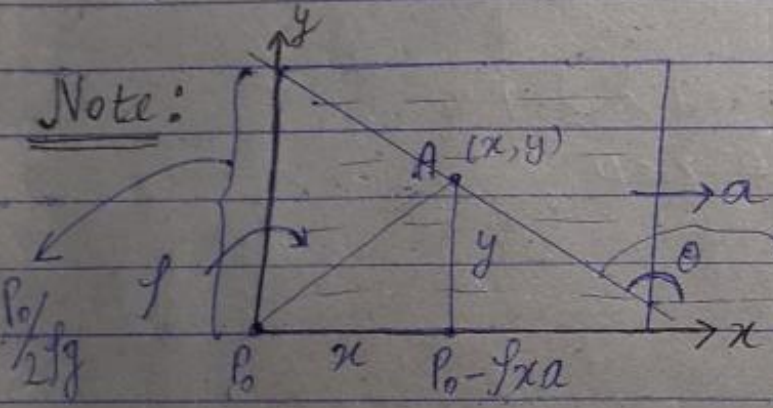


$P_x A - P_y A = (\rho l A)\, a$

$P_x - P_y = \rho l a$

if we move opposite to acceleration, pressure will increase.

$\frac{P_x - P_0}{l}$ = Pressure Gradient = ρa



Note:

$P_A = P_0 - \rho x a - \rho g y$

→ equipressure surface

find the relation between coordinates x and y at which pressure is $\frac{P_0}{2}$

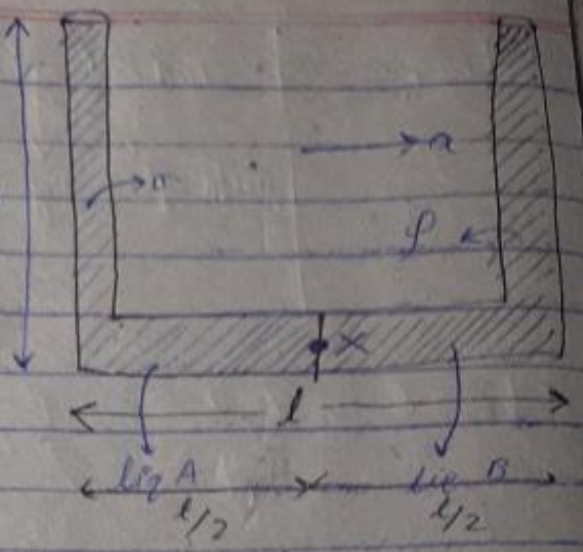
$P = P_0 - \rho x a - \rho g y = \frac{P_0}{2}$

$\frac{P_0}{2} = \rho(xa + gy)$

$xa - gy = \frac{P_0}{2\rho} \Rightarrow \boxed{y = -\frac{x}{g}a + \frac{P_0}{2\rho g}}$

Patm

(Q) Liquids are in hydrostatic condition wrt tube. Find acceleration.



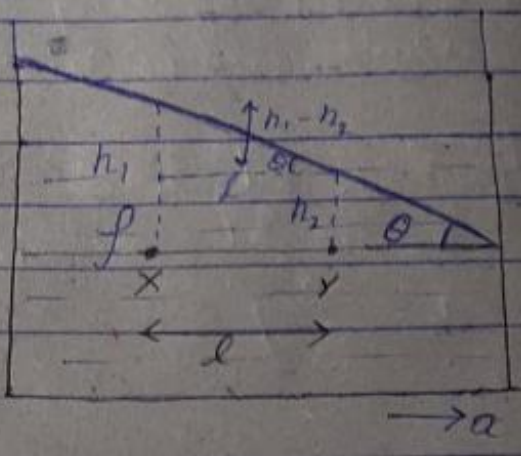
$$P_x = P_{atm} + \sigma g H - \sigma a \frac{l}{2}$$

$$P_x = P_{atm} + \rho g H + \rho a \frac{l}{2}$$

$$\Rightarrow (\sigma - \rho) g H = (\rho + \sigma) a \frac{l}{2}$$

$$\boxed{a = \frac{2(\sigma - \rho) g H}{(\rho + \sigma) l}}$$

(Q) Find angle θ



$$\tan\theta = \frac{h_1 - h_2}{l} \quad \text{— ①}$$

$$P_x - P_y = \rho l a \quad \text{— ②}$$

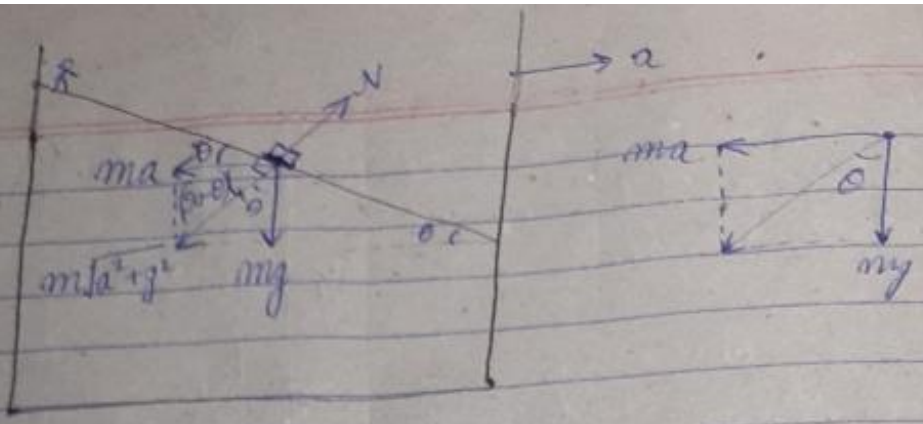
$$P_x = P_{atm} + \rho g h_1, \quad P_x - P_y = \rho g (h_1 - h_2) \quad \text{— ③}$$

$$P_y = P_{atm} + \rho g h_2$$

$$\Rightarrow \rho l a = \rho g (h_1 - h_2) \Rightarrow \frac{h_1 - h_2}{l} = \frac{a}{g} = \tan\theta$$

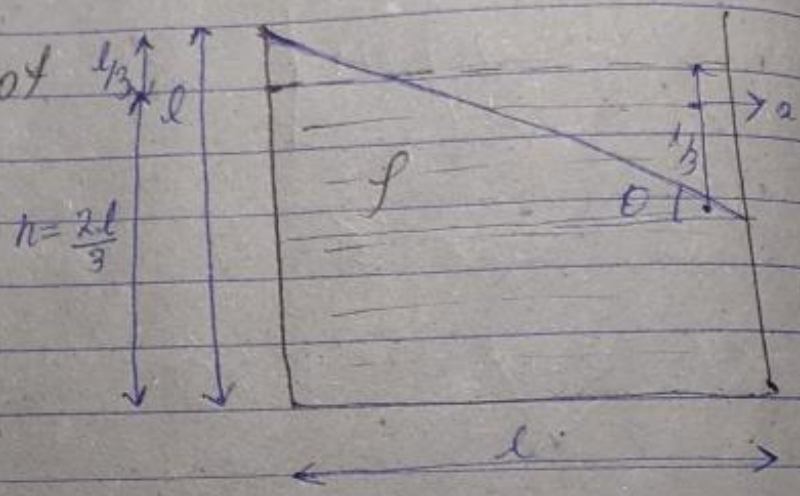
$$\therefore \boxed{\tan\theta = \frac{a}{g}}$$

Also:



$$\tan\theta = \frac{ma}{mg} = \frac{a}{g}$$

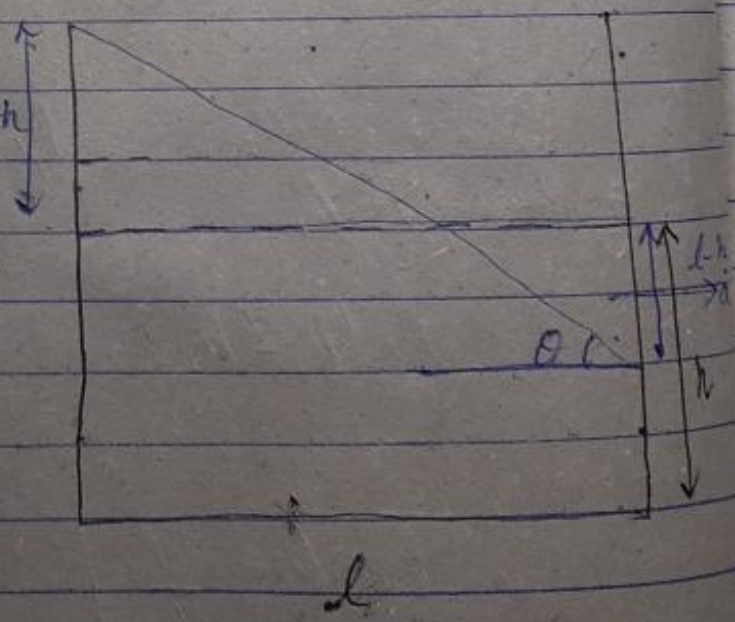
(Q) Find max. value of a without spilling water from container.



$$\tan\theta = \frac{2l}{3l} = \frac{2}{3} = \frac{a}{g}$$

$$a = \frac{2}{3}g$$

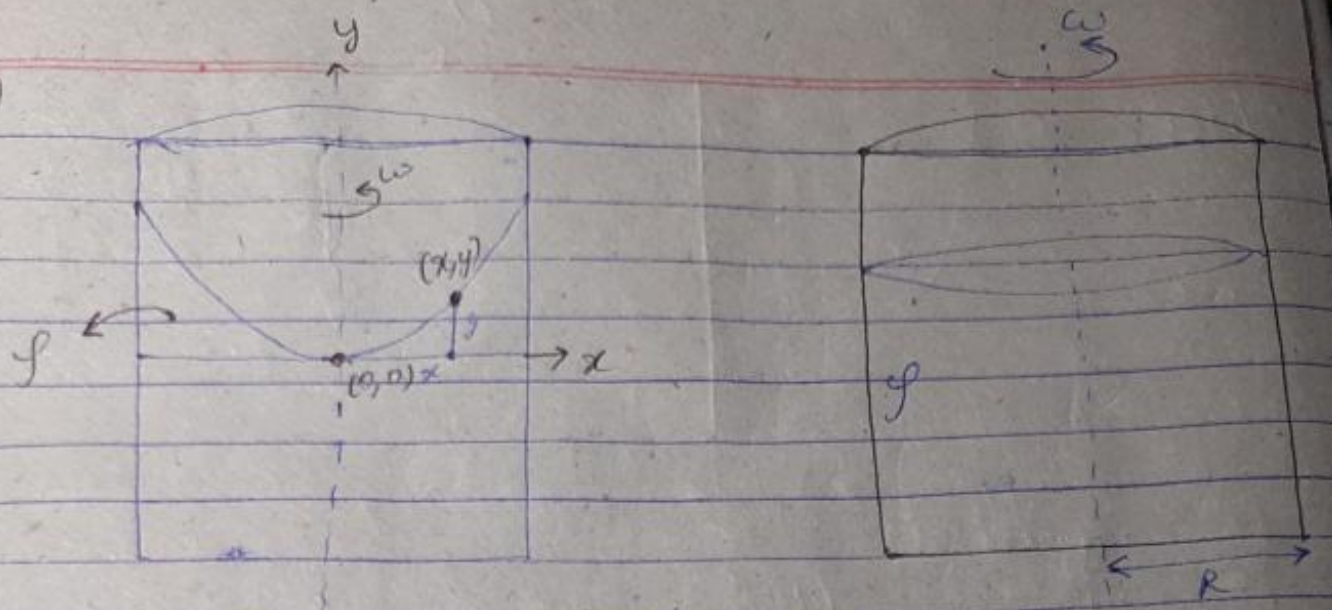
(Q) Find max value of a without spilling liquid from container.



$$\tan\theta = \frac{2(l-h)}{l} = \frac{a}{g}$$

$$a = \frac{2g(l-h)}{l}$$

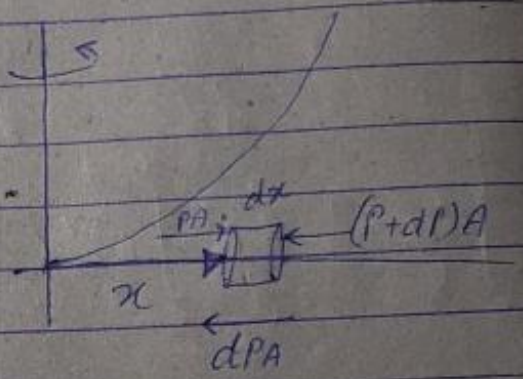
(1)



Find relation between (x, y) and free surface.

$$P = P_{atm} + \rho g y \quad \text{——— (1)}$$

$$a_c = \omega^2 x$$



$$\therefore dPA = (\rho A dx)\omega^2 x$$

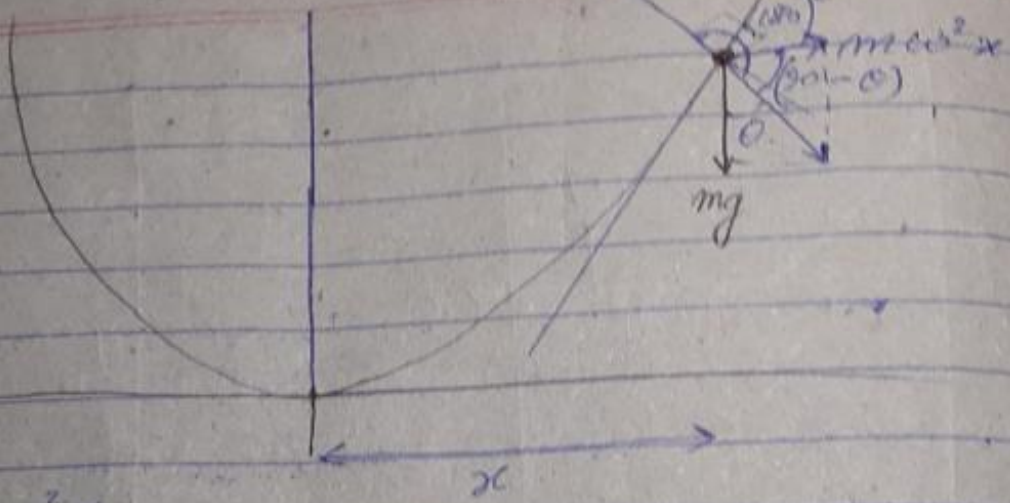
$$\int_{P_{atm}}^{P} dP = \rho\omega^2 \int_0^x x\,dx$$

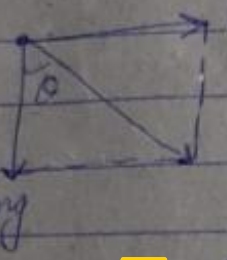
$$\left|P\right|_{P_{atm}}^{P} = \rho\omega^2 \left|\frac{x^2}{2}\right|_0^x \Rightarrow P = P_{atm} + \frac{\rho\omega^2 x^2}{2} \quad \text{——— (2)}$$

$$\therefore \rho g y = \frac{\rho\omega^2 x^2}{2} \Rightarrow gy = \frac{\omega^2 x^2}{2}$$

$$\boxed{y = \frac{\omega^2 x^2}{2g}}$$

Aliter:



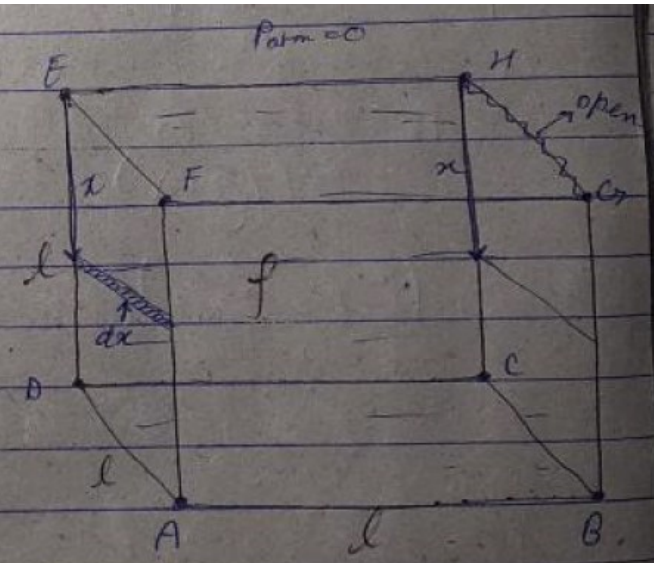


$$\tan\theta = \frac{m\omega^2 x}{mg} = \frac{\omega^2 x}{g} = \frac{a}{g}$$

$$\frac{dy}{dx} = \frac{\omega^2}{g} x$$

$$\int_0^y dy = \frac{\omega^2}{g} \int_0^x x\,dx$$

(8) (i) Find pressure points A, B, C, D, E, F, G, H



$P_G = P_H = 0 = P_E = P_F$ as on same hor. level.

$P_A = P_B = P_C = P_D = \rho g l$

(ii) Find force on each face of cube.

$P_{ABCD} = \rho g l^3 = mg$

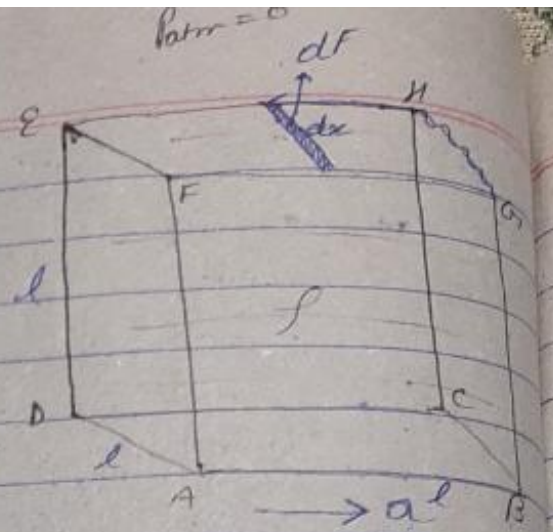
$P_{EFGH} = 0$

$dF = \rho g x l dx$

$F = \int dF = \int_0^l \rho g x l dx$

$F = \rho g l \int_0^l x dx = \frac{\rho g l^3}{2}$

Gravity free space.



(i) P at each point.

$P_G = P_H = P_E = P_F = 0$

$P_G = P_H = P_C = P_B = 0$

$P_E = P_F = P_D = P_A = \rho l a$

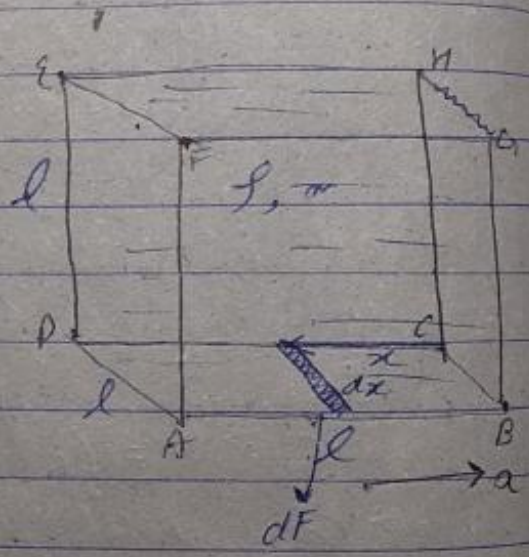
(ii) Find force on each face.

$F_{BCHG} = 0$, $F_{ADEF} = \rho l^3 a = ma$

$$dF = (\rho x a) l\, dx \Rightarrow F = \int_0^F dF = \rho a l \int_0^l x\, dx = \frac{\rho l^3 a}{2}$$

$= F_{EFGH}$

$F_{ABCD} = \rho l^3 a/2$



(iii) $P_G = P_H = 0$

$P_E = F_F = \rho l a$

$P_B = P_C = \rho g l$

$P_A = P_D = \rho g l + \rho l a$

$= \rho l (g + a)$.

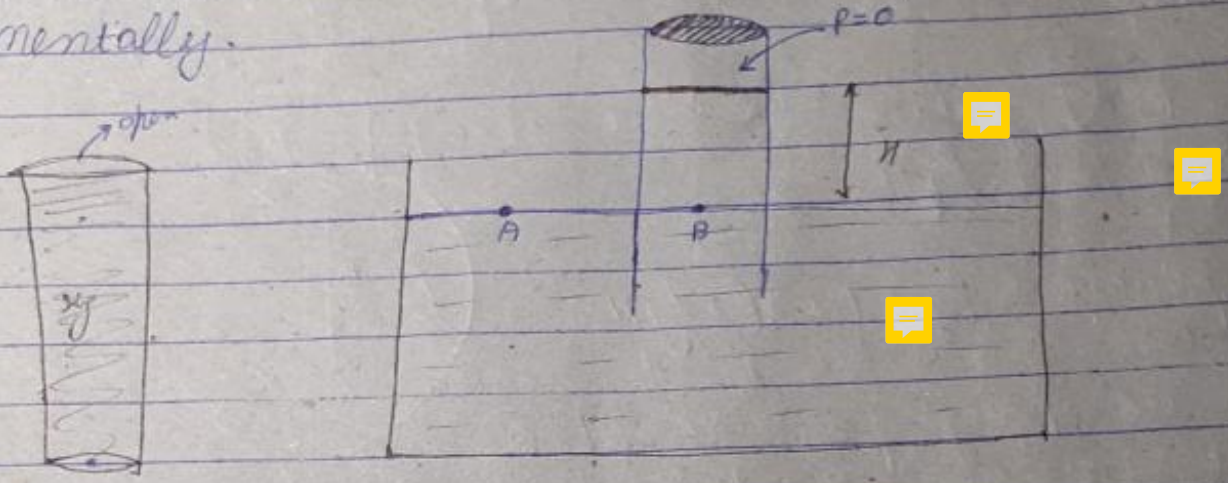
(iii) $dF = (\rho g l + \rho x a) l\, dx$

$$F = \int_0^F dF = \rho g l^2 \int_0^l dx + \rho a l \int_0^l x\, dx$$

$$\Rightarrow F = \rho g l^3 + \frac{\rho a l^3}{2} = mg + \frac{ma}{2} = \frac{3}{2} mg$$

• Barometer (Torricelli)

It is used to find atmospheric pressure experimentally.



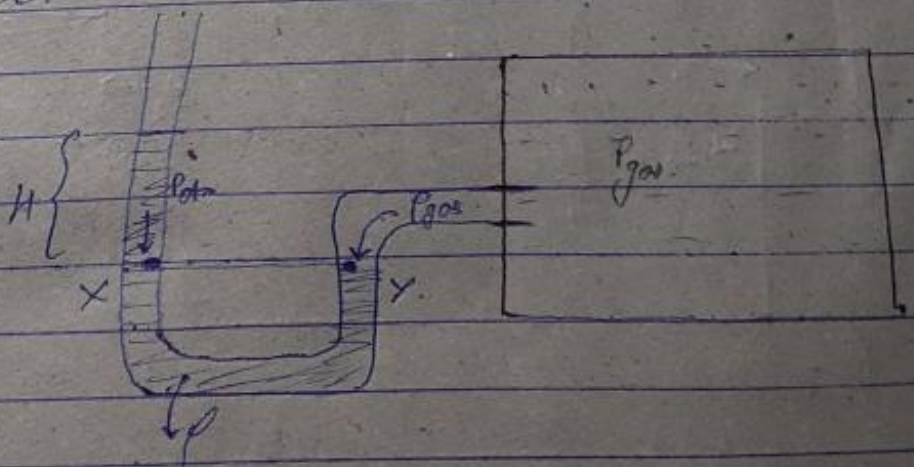
$P_A = P_B \Rightarrow P_{atm} = 0 + \rho g H \Rightarrow P_{atm} = 13600 \times 9.8 \times 0.76.$

$P_{atm} = 1.01 \times 10^5$ Pa

• Manometer

It is used to measure pressure of gas in a closed container.

at equilibrium.



$P_x = P_y$

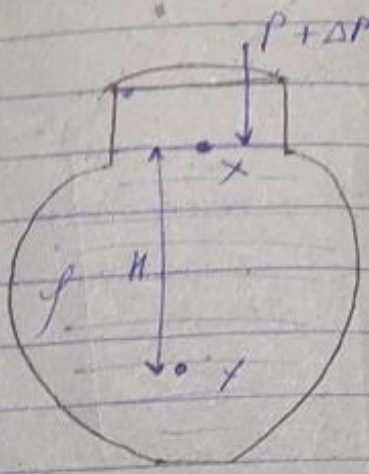
$P_{gas} = P_{atm} + \rho g H$

$P_{gas} - P_{atm} = \rho g H$

$P_{excess} = \rho g H$ = gauge pressure

• Pascal's Law

It was based on the property of incompressibility of fluid.



$P_y - P_x = \rho g H$.

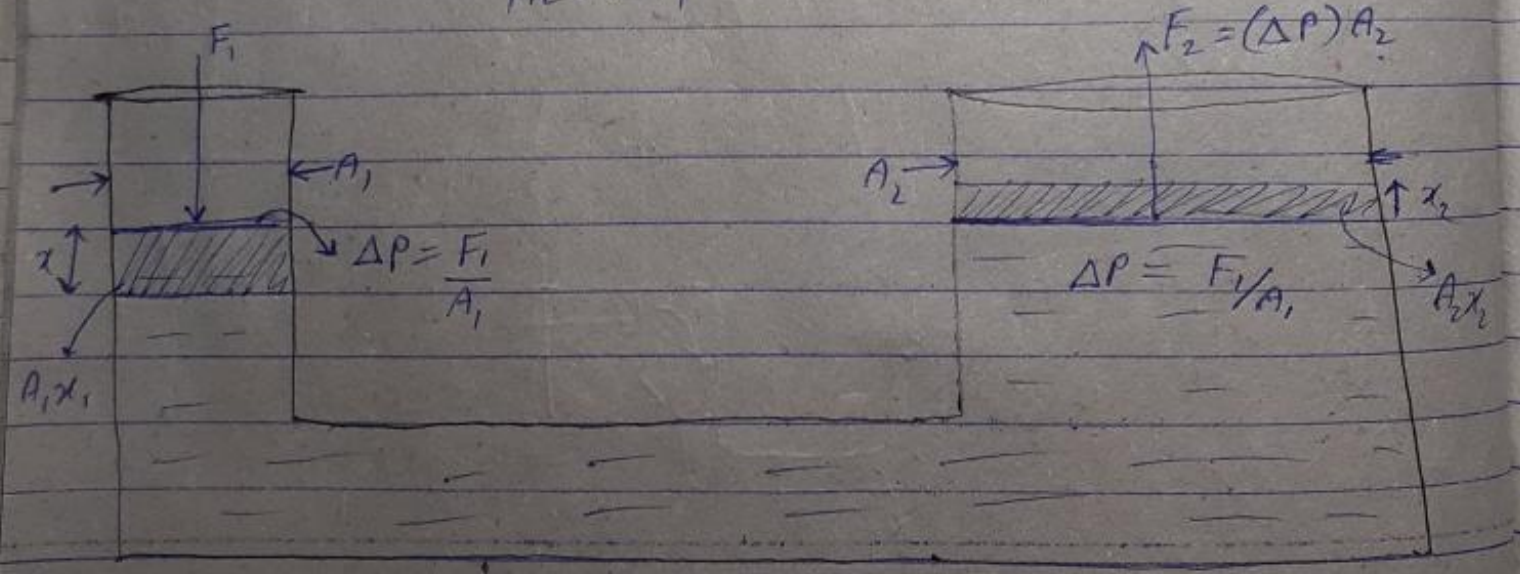
$P_y = P_x + \rho g H$

new pressure $\Rightarrow P_y = P + \Delta P + \rho g H$

- If we change pressure at a point in fluid, this change in pressure will be transmitted to the entire connected liquid.

This property is used in hydraulic lift.

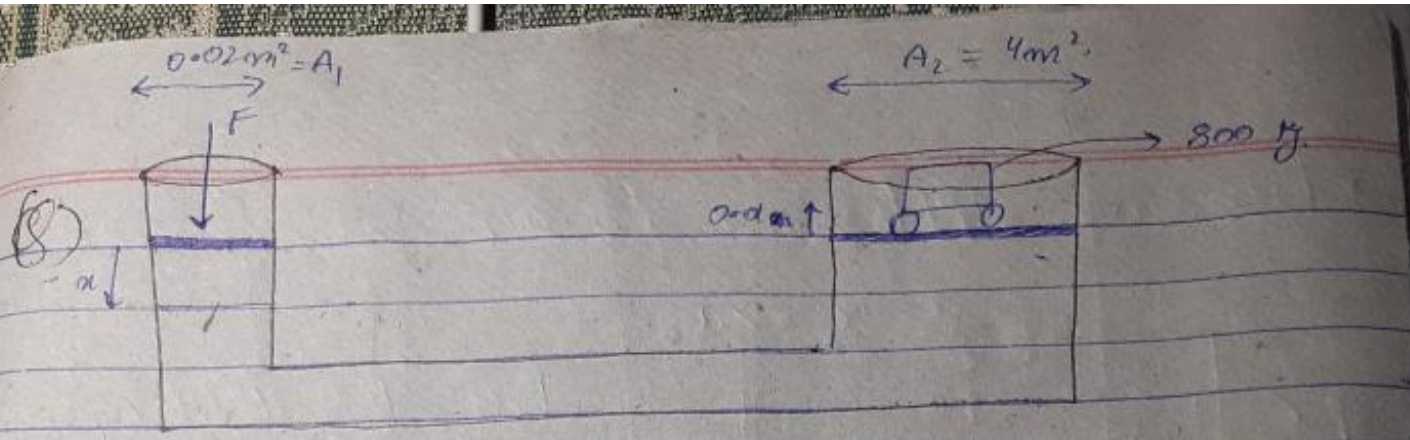
$A_2 >> A_1$



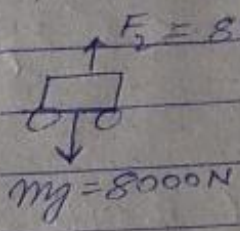
$F_2 = F_1 \frac{A_2}{A_1} \Rightarrow F_2 >> F_1$

$A_1 x_1 = A_2 x_2$; $W_1 = F_1 x_1 = \Delta P A_1 x_1$

$W_2 = f_2 x_2 = \Delta P A_2 x_2$



How much force to be applied on A_1 to lift car slowly.



$$\therefore \frac{F}{0.02} = \frac{8000}{4}$$

$$F = 2000 \times \frac{2}{100} = 40\ N$$

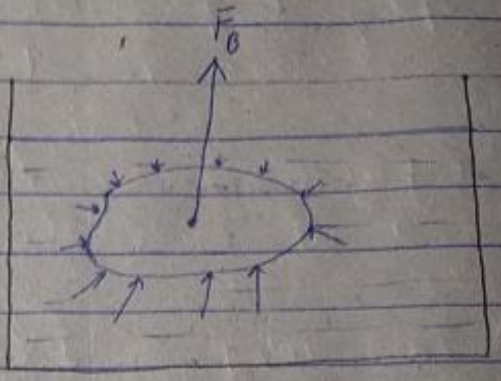
If car is to be moved 0.1 m then find work done.

$$\Rightarrow 0.02 \times x = 4 \times 0.1$$

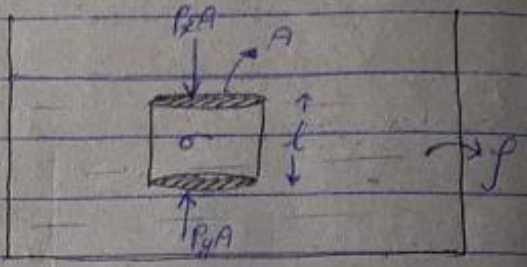
$$x = \frac{40}{2} = 2$$

- **Buoyancy / Upthrust force**

contact force applied by fluid on solid in vertically upward direction due to pressure difference when container is at rest or unaccelerated.



(8) Find force applied by liquid on submerged solid.



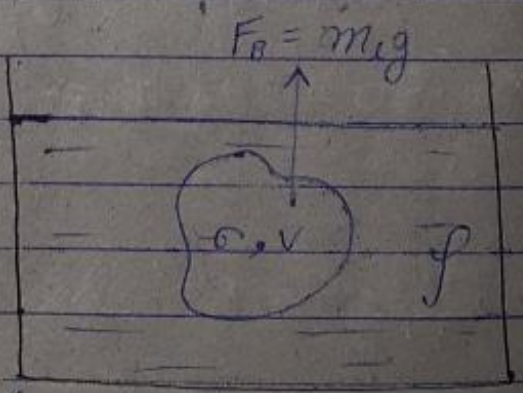
$F_{net} = P_yA - P_xA = (P_y - P_x)A$

$F_{net} = \rho g l A = F_B$

$F_B = (\rho l A)g = \rho_{liq} V_{solid} g = m_L g$

where m_L is mass of liquid displaced by solid.

- **Archimedes Principle**



- $F_B = \rho V g$

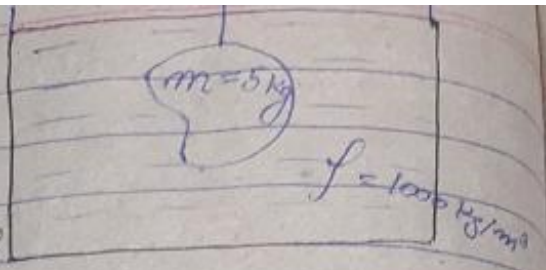
It states that buoyancy force applied by the liquid is equal to weight of displaced liquid.

If $a\uparrow \Rightarrow F_B = m_L g_{eff} = \rho V(g+a)$

If $a\downarrow \Rightarrow F_B = \rho V(g-a)$



(8) specific gravity of solid is 4. then find,



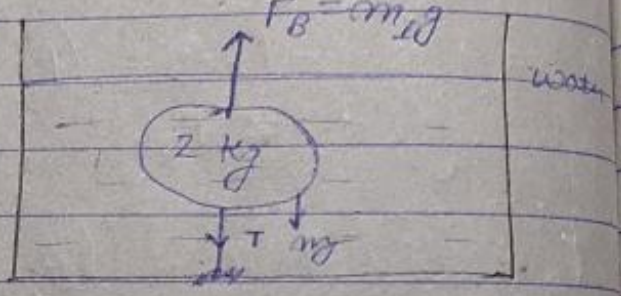
① ~~Bgoy~~ Buoyancy force of solid
② a of solid.

① $F_B = m_L g = \rho V g = \frac{\rho m g}{\sigma} = \frac{mg}{(\sigma/\rho)} = \frac{mg}{\text{sp. gravity}}$

$F_B = \frac{5 \times 10}{4} = 12.5\ N$

② $50 - 12.5 = 5a \Rightarrow a = \frac{37.5}{5} = 7.5\ m/s^2$

(8) specific gravity of solid = 0.4
find; ① Buoyancy force on the solid.
② Tension in string.

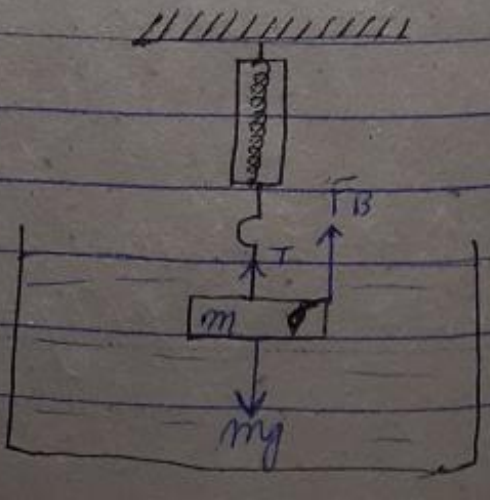


① $F_B = m_L g = \rho V g = \frac{\rho m g}{\sigma} = \frac{mg}{[5 \cdot g]} = 2.5mg = 50\ N$

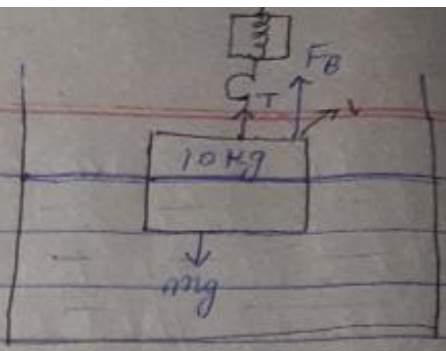
② $50 = 20 + T \Rightarrow T = 30\ N$

- <u>Apparent weight in liquid.</u>

$T + F_B = mg$
$T = mg - F_B$
$T = mg - m_L g$
$T = mg - \frac{\rho m g}{\sigma}$
$T = mg\left(1 - \frac{\rho}{\sigma}\right)$



(5) density of solid = 5000 kg/m³
density of liquid = 2000 kg/m³
volume submerged = $\frac{3}{5}V$



Find reading of spring balance.

$T + F_B = mg \Rightarrow T + m_l g = mg$

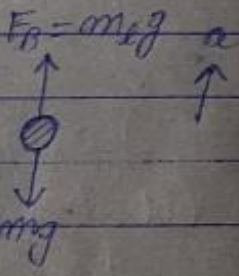
$T + \rho V_i g = mg \Rightarrow T + \rho g \frac{3}{5}\frac{m}{\rho} = mg$

$T + 2000 \times 10 \times \frac{3}{5} \times \frac{10}{5000} = 100$

$T + 24 = 100 \Rightarrow T = 100 - 24 = 76$ N

∴ Reading of balance = 7.6 kgf

(6) A very small ball of sp. gravity 0.4 is below the free surface of water by 20cm. If the ball is released find;
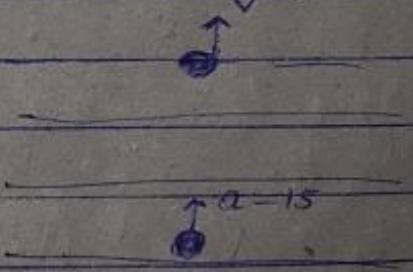


① a of ball just after release.

$F_B = \frac{mg}{\rho g} = \frac{mg}{0.4} = 2.5mg$

$2.5mg - mg = ma \Rightarrow a = 15 m/s^2$

② find speed of ball when it reaches the free surface.



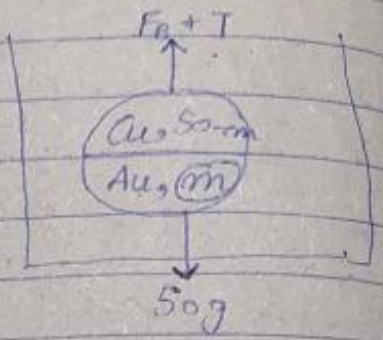
$v^2 = u^2 + 2as \Rightarrow v^2 = 2 \times 15 \times 0.2$

$v = \sqrt{6}$ m/s.

③ find max height achieved by ball from free surface. $v_f = 0 \Rightarrow 0^2 = 6 - 2 \times 10 \times H$

$H = \frac{6}{20} = 30$ cm

(Q) An ornament is made of Cu, Au of specific gravity 10 and 20 respectively. The actual mass of ornament is 50 gm when weighed in water then reading becomes 46 gm. Find mass of gold in ornament.



$\Rightarrow T + F_B = mg$

$F_B = 50g - 46g = 4g$

$\Rightarrow V\rho g = 4g = F_{B1} + F_{B2}$

$\Rightarrow 4g = \frac{(50-m)g}{10} + \frac{mg}{20}$

$\Rightarrow 4 = \frac{100 - 2m + m}{20} \Rightarrow m = 20gm$

(Q) A copper block is having a cavity. mass of Cu block is 260 gm and sp. gravity = 10. When this Cu block is weighted in water reading of spring balance becomes 225 gm. Find volume of cavity.



$225g + F_B = 260g$

$F_B = 35g$

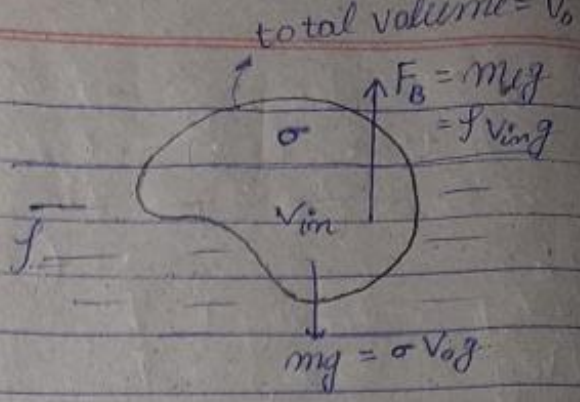
$F_B = V\rho g = (V_{Cu} + V_0)\rho g$

$F_B = \frac{m}{\sigma}\rho g + V_0\rho g$

$F_B = \frac{mg}{\text{Sp. gravity}} + V_0\rho g \Rightarrow \frac{260\rho}{10} + V_0\rho g = 35g$

$V_0 = \frac{9}{\rho} = \frac{9}{1} = 9\,cm^3$

• Floatation



For eq. $F_B = mg$

$\Rightarrow m_l g = mg$

$\rho V_{in} g = \sigma V_0 g$

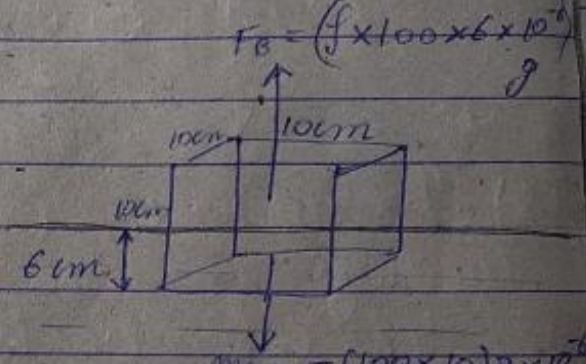
$\frac{V_{in}}{V_0} = \frac{\sigma}{\rho}$ = $\frac{\text{density of solid}}{\text{density of liquid}}$

$V_{in} < V_0 \Rightarrow \sigma < \rho$ (for floatation)

Fraction of volume of solid submerged

$= \frac{V_{in}}{V_0} = \frac{\sigma}{\rho}$

(Q) Find density of cube.



$(\rho \times 100 \times 6 \times 10^{-6}) g = (\sigma \times 100 \times 10 \times g \times 10^{-6})$

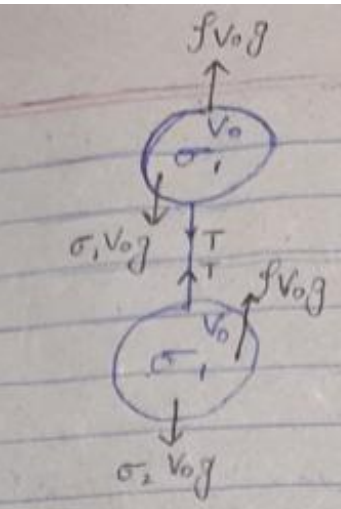
$\Rightarrow 6\rho = 10\sigma \Rightarrow \sigma = 900 \text{ kg/m}^3$ $\quad \rho = 1500 \text{ kg/m}^3$

(or) $\frac{V_{in}}{V_0} = \frac{\sigma}{\rho}$

$\Rightarrow \frac{100 \times 10^{-4} \times 6 \times 10^{-2}}{100 \times 10^{-4} \times 10 \times 10^{-2}} = \frac{\sigma}{1500}$

$\sigma = 900 \text{ kg/m}^3$

(Q) Find density of liquid such that both objects remain in equilibrium



(I)

$\Rightarrow \rho V_0 g = \sigma_1 V_0 g + T$

$\rho V_0 g = \sigma_2 V_0 g - T$

$2\rho V_0 g = (\sigma_1 + \sigma_2) V_0 g$

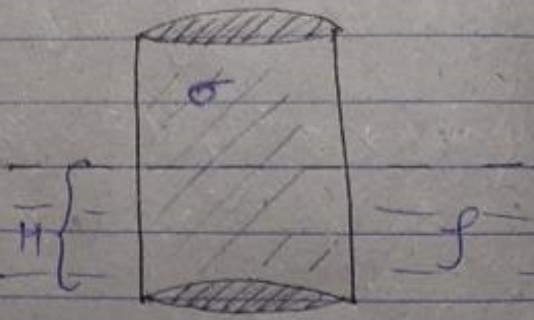
$\rho = \frac{\sigma_1 + \sigma_2}{2}$

(II) NLM on whole system

$\Rightarrow 2\rho V_0 g = \sigma_1 V_0 g + \sigma_2 V_0 g$

$\rho = \frac{\sigma_1 + \sigma_2}{2}$

(Q) This cylinder is floating in liquid. If the cylinder is displaced downwards prove that it performs SHM and also find Time period of this SHM.

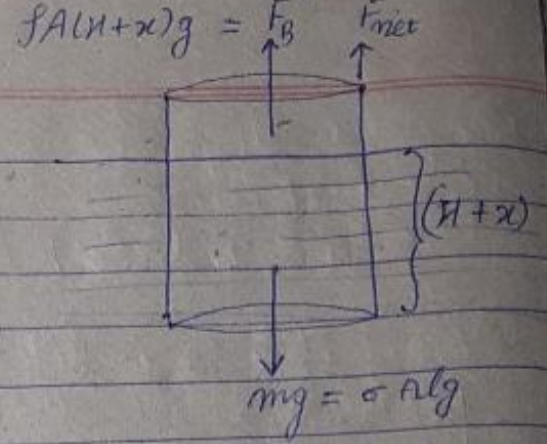


let σ, ρ be densities of cylider, liq and length of cylinder be l.

$\rho A H g = \sigma A l g$

$\rho H = \sigma l \quad \text{---} (1)$





$F_{net} = F_B - mg$

$= \rho A(H+x)g - \sigma Alg$

$= \rho AHg + \rho Axg - \sigma Alg$

by (1) $\rho AHg = \sigma Alg$

$\therefore F_{net} = (\rho Ag)x \Rightarrow F_{net} \propto x \Rightarrow$ SHM Proved.

$\Rightarrow (\rho Ag)x = (\sigma Al)a \Rightarrow a = \left(\frac{\rho g}{\sigma l}\right)x$

$\therefore \omega = \sqrt{\frac{\rho g}{\sigma l}} = \sqrt{\frac{\rho g}{\rho H}}$ (by (1)) $\Rightarrow \omega = \sqrt{\frac{g}{H}}$

$\therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{H}{g}}$

- **Centre of Buoyancy**

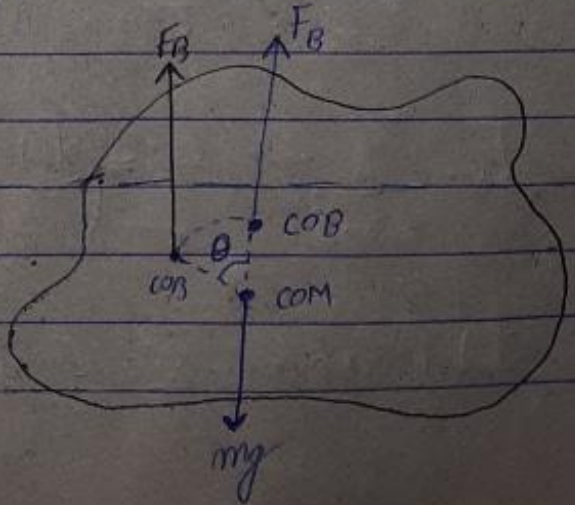
It is a point at which we can assume the buoyancy force to be acting

It is a centre of submerged part.

(1) angular displacement of body anticlockwise.

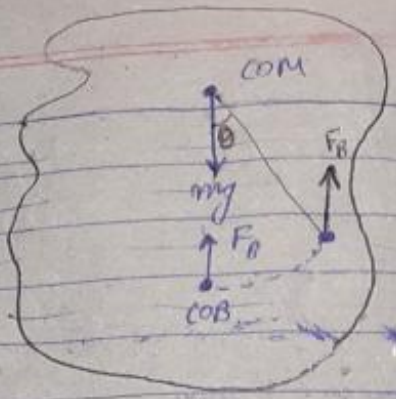
τ_{net} is clockwise.

Stable eq. position.



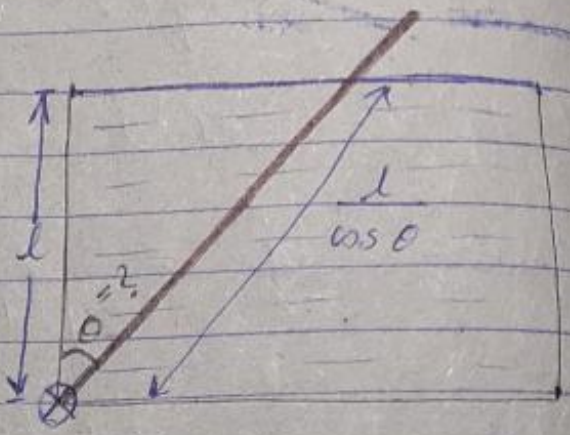
Angular displacement anticlockwise.

τ_{net} = anticlockwise

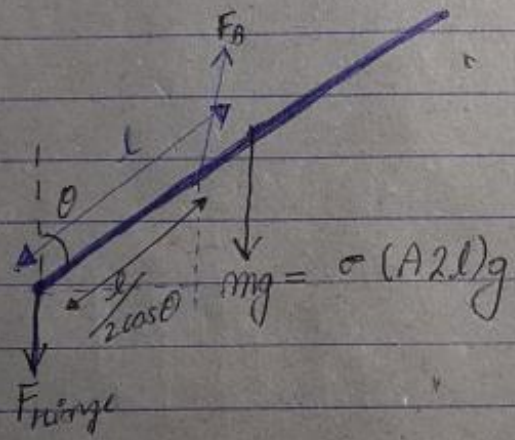


Unstable equilibrium position.

(Q) Rod of length $2l$ and specific gravity = 0.5 is hinged from corner



① Draw FBD of Rod.



② Find the angle θ

since rod is at equilibrium. ∴ $\tau_{net} = 0$

$$\tau_{F_B} = \tau_g$$

$$\Rightarrow F_B \cdot \frac{l}{2\cos\theta} \sin\theta = \left(\frac{\rho A l}{\cos\theta}\right) g \cdot \frac{l \sin\theta}{2\cos\theta}$$

$$\Rightarrow \frac{\rho A l}{2} \sec\theta \cdot \frac{\sin\theta}{\cos\theta} g = (\sigma A 2l g)\, l \sin\theta$$

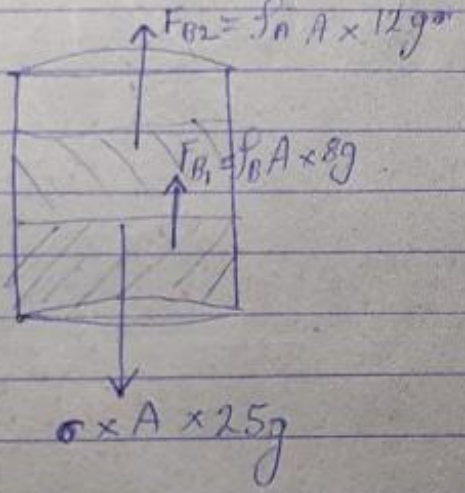
$$\Rightarrow \frac{\rho}{2}\sec^2\theta = 2\sigma \Rightarrow \sin\theta = 0$$
$$\theta = 0$$

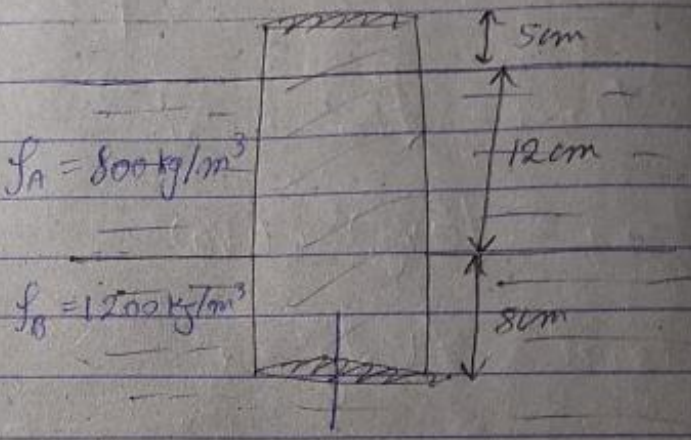
$$\sec^2\theta = \frac{4\sigma}{\rho} = 4 \times 0.5 = 2$$

$$\sec^2\theta = 2 \Rightarrow \sec\theta = \sqrt{2}$$
$$\theta = 45^\circ$$

$\therefore \theta = 0^\circ$ (or) 45°

(Q) Find density of solid cylinder.



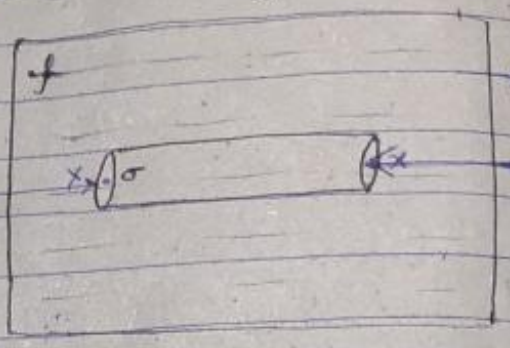


$$\Rightarrow 1200 \times A \times 8 \times g + 800 \times A \times 12 g = \sigma \times A \times 25g$$
$$= \sigma A 25g$$

$$\sigma = \frac{1200 \times 8 + 800 \times 12}{25} = 768 \ kg/m^3$$

(a) Buoyancy force in accelerating container

$P_y - P_x = \rho a l$



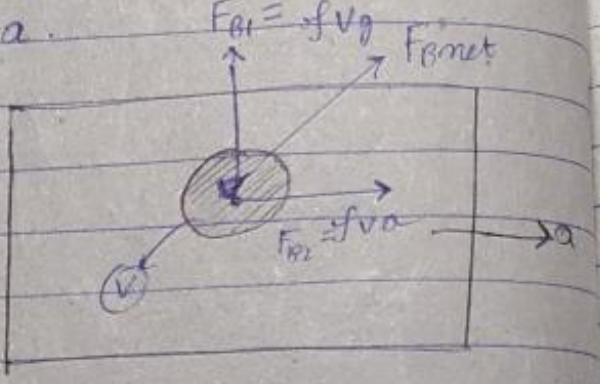
$F_{net} = F_1 - F_2$

$= (P_y - P_x) A$

$= P_y A - P_x A$

$= (P_y - P_x) A$

$= \rho a l A = (\rho l A) a = m_f a$



$F_{Bnet} = \sqrt{F_{Bi}^2 + F_{Bi}^2}$

$F_{Bnet} = \sqrt{(\rho V g)^2 + (\rho V a)^2}$

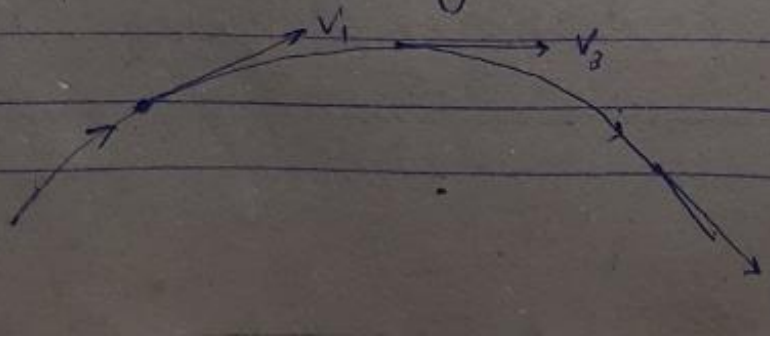
$F_{Bnet} = \rho V \sqrt{g^2 + a^2}$

$= \rho V g_{eff}$

- **Dinamycs of Fluids**

Fluid is in motion.

Streamline: It is an imaginary line or curve along which fluid particles are moving.



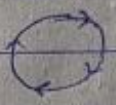
Streamline is a line or curve, tangent to which provides the direction of velocity of fluid particle at that point.

2 streamlines will not intersect.

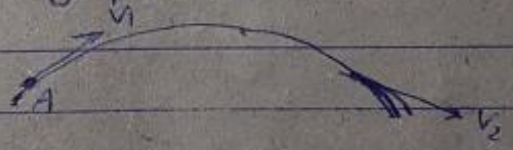
Assumptions :-

(i) density of fluid is independent of pressure applied.

(ii) Fluid is non viscous.

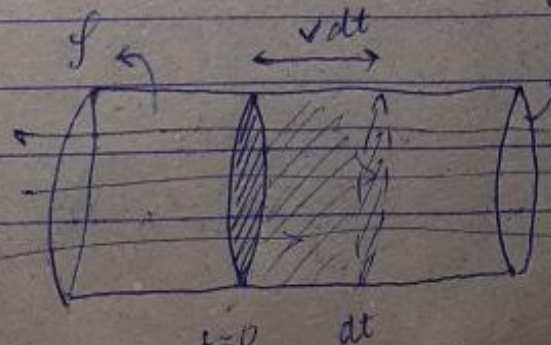
(iii) Fluids are irrotational, ie there will be no rotational motion. 

(iv) Fluid particles will move on fixed paths and these are known as stream lines. Velocity of any fluid particle on any point on stream line remains same.



- Equation of Continuity

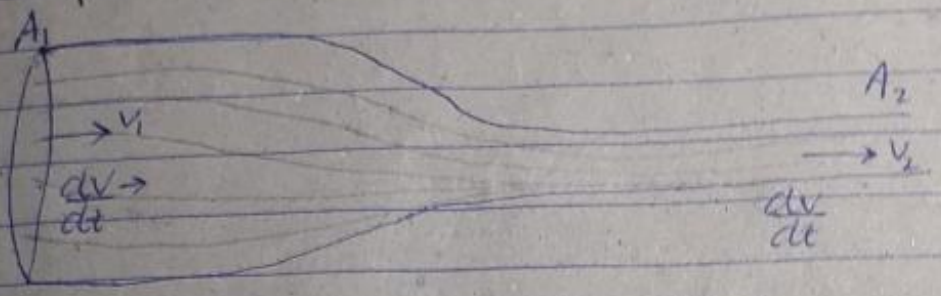
Volume flow rate : Volume of fluid flowing in a tube per unit time.



$$dV = A\,v\,dt$$

$$\frac{dV}{dt} = Av$$

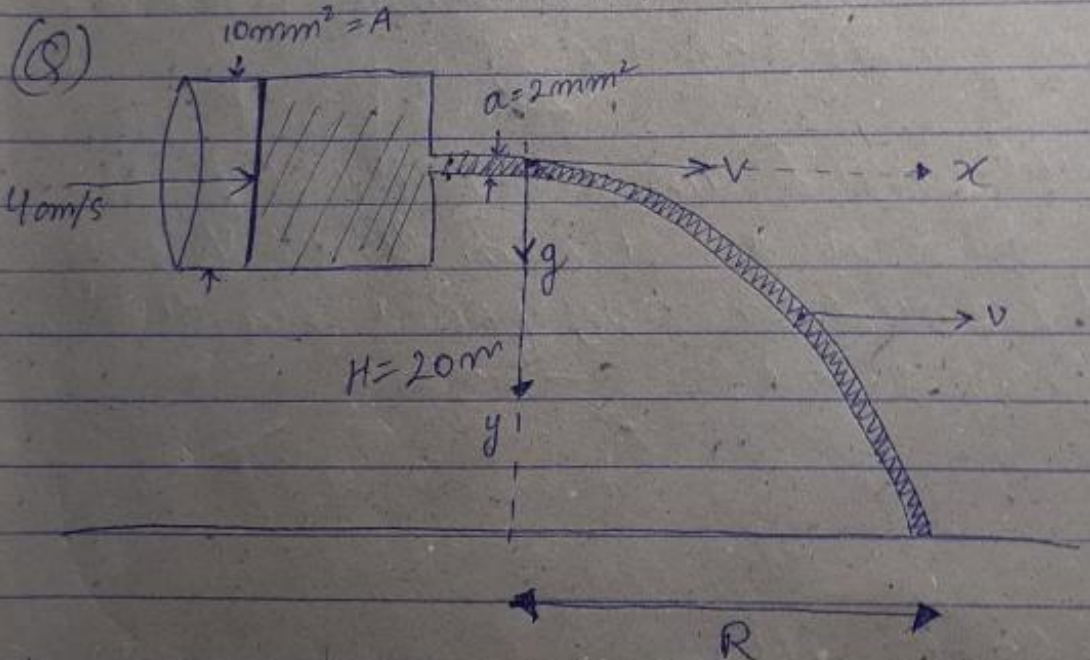
• Mass flow rate is rate at which mass is flowing $= \frac{dm}{dt} = \frac{\rho dV}{dt} = \rho A v$



$$\left(\frac{dV}{dt}\right)_1 = \left(\frac{dV}{dt}\right)_2 \Rightarrow \boxed{A_1 v_1 = A_2 v_2} \qquad v \propto \frac{1}{A}$$

Streamline density ∝ velocity of fluid

(Q)



Find time taken by fluid to reach ground after ejected from pipe also find range R.

$u_y = 0$, $a_y = g$, $S_y = H = 20m$

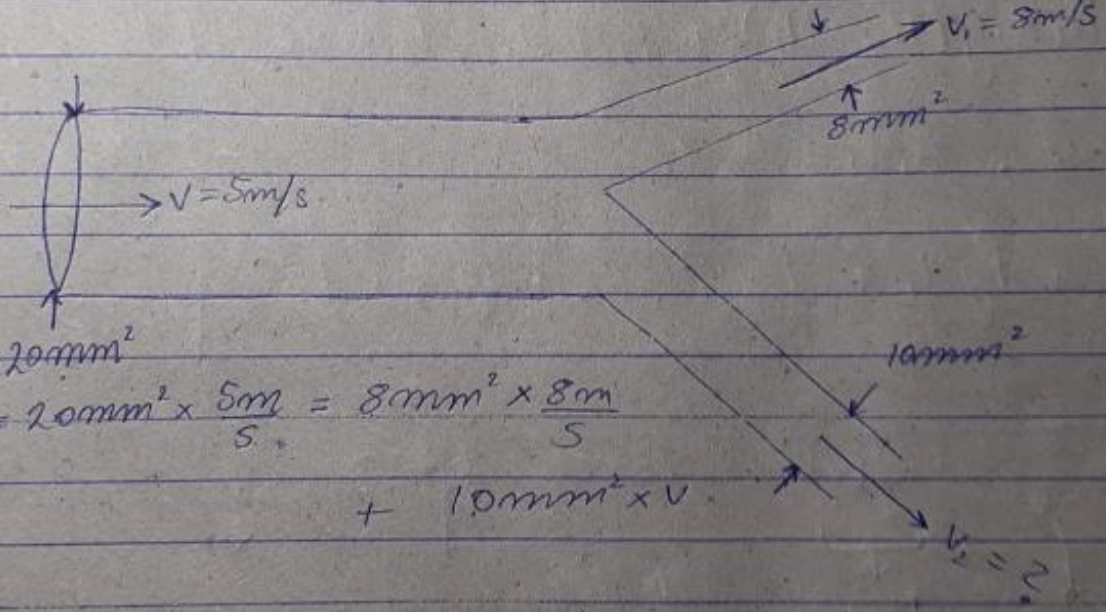
$$S_y = u_y t + \frac{1}{2} a_y t^2 \Rightarrow t = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2\times 20}{10}} = 2s$$

now, $10mm^2 \times \frac{4cm}{s} = 2mm^2 \times V$

$V = 20\ cm/s = 0.2\ m/s$.

$$x = u_x t + \frac{1}{2} a_x t^2 = R = 0.2 \times 2 = 0.4m$$

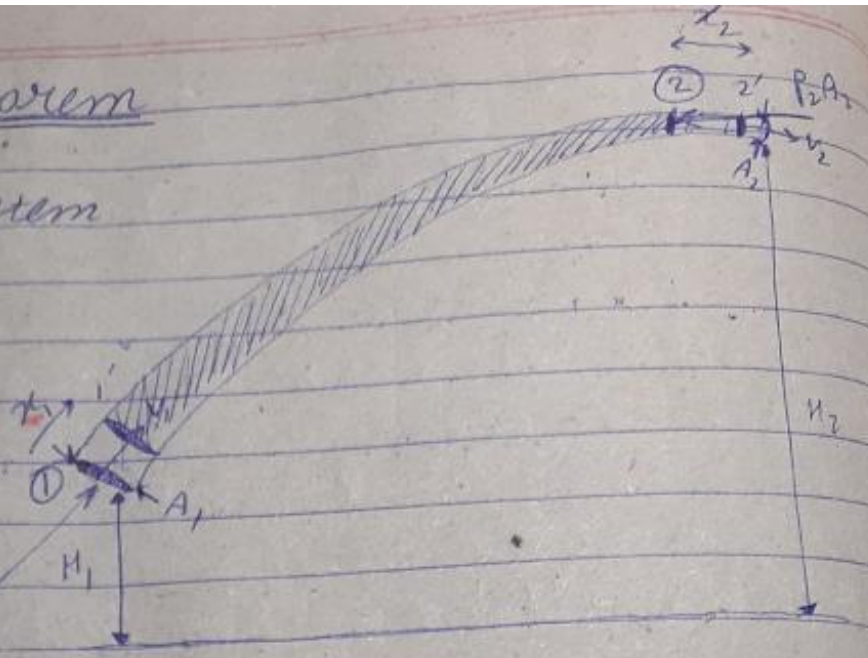
(a)



$$\frac{dV}{dt} = 20mm^2 \times \frac{5m}{s} = 8mm^2 \times \frac{8m}{s} + 10mm^2 \times V$$

$$\Rightarrow \frac{100 - 64m}{10} = V_2 \Rightarrow V_2 = 8.6m/s$$

• Bernoulli's Theorem



① and ② is a system

$W_g + W_p = K_f - K_i$

$\Rightarrow -\rho A_1 x_1 g(H_2 - H_1) + P_1 A_1 x_1 - P_2 A_2 x_2 = \frac{1}{2}\rho A_2 x_2 v_2^2 - \frac{1}{2}\rho A_1 x_1 v_1^2$

Since, $A_1 x_1 = A_2 x_2$

$$\Rightarrow \boxed{P_1 + \rho g H_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g H_2 + \frac{1}{2}\rho v_2^2}$$

$$\boxed{P + \rho g H + \frac{1}{2}\rho v^2 = \text{constant}}$$

- $\frac{1}{2}\rho v^2$ → K per unit volume or K density
- $\rho g H$ → U density
- P → Pressure energy density

$$\frac{P}{\rho g} + H + \frac{v^2}{2g} = \text{constant.}$$

where H is called gravitational head and $\frac{P}{\rho g}$ is called Pressure head.

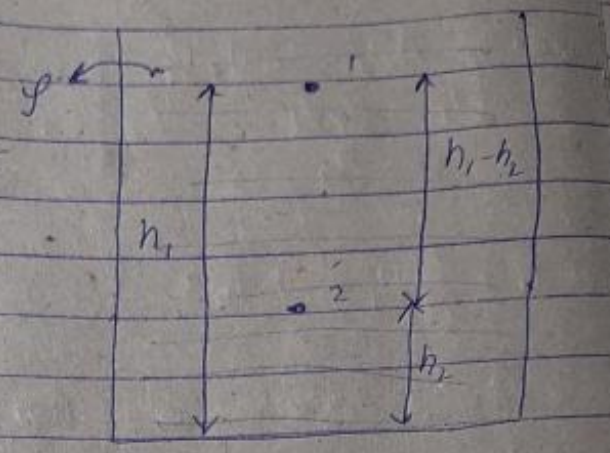
Hydrostatic condition is a special case of Bernoulli's theorem.

$$P + \rho g H + \frac{1}{2}\rho v^2 = \text{const.}$$

$$P_1 + \rho g H_1 + \frac{1}{2}\rho v^2 = P_2 + \rho g H_2 + \frac{1}{2}\rho v^2$$

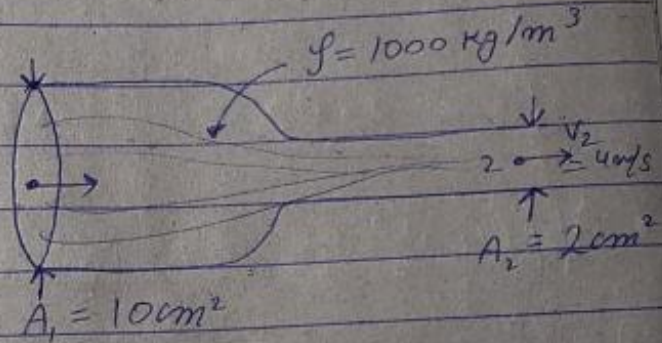
$$P_2 - P_1 = \rho g (H_1 - H_2)$$

$$P_2 - P_1 = \rho g H$$



(8) $P_2 = 3 \times 10^5\ N/m^2$

Find speed of fluid and Pressure at ①



Sol: ① $A_1V_1 = A_2V_2 \Rightarrow 10cm^2 \times V_1 = 2cm^2 \times 4\ m/s$

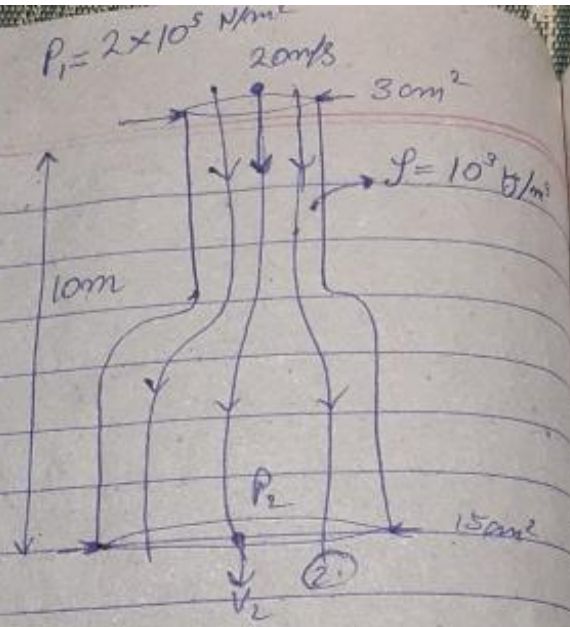
$V_1 = 0.8\ m/s$

② $P_1 + \rho g H_1 + \frac{1}{2}\rho V_1^2 = P_2 + \rho g H_2 + \frac{1}{2}\rho V_2^2$

$$\Rightarrow P_1 + \frac{1}{2} \times 1000 \times 0.64 = 3 \times 10^5 + \frac{1}{2} \times 1000 \times 16$$

$$P_1 = (3 \times 10^5 + 7.68 \times 10^3)\ N/m^2$$

(Q) Find speed of fluid and pressure at ②



① $A_1V_1 = A_2V_2$

$V_2 = \frac{A_1V_1}{A_2} = 4$ m/s.

② $P_1 + \rho g H_1 + \frac{1}{2}\rho V_1^2 = \text{const.}$

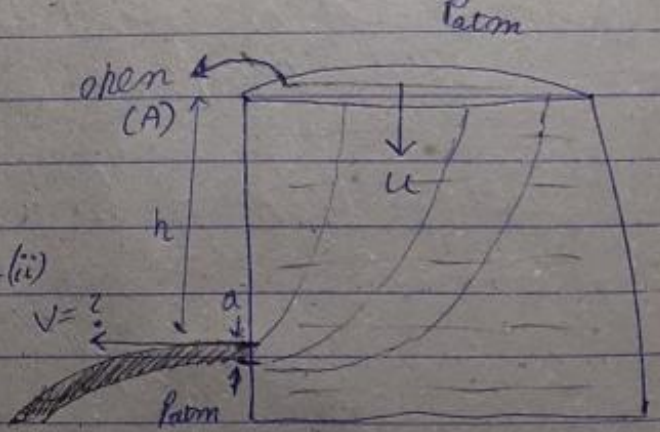
$\Rightarrow 2\times10^5 + 10^3\times10\times10 + \frac{1}{2}\times10^3\times(20)^2 = P_2 + 10^3\times10\times0 + \frac{1}{2}\times10^3\times16$

$P_2 = 2.92\times10^5$ N/m²

• Applications of Bernoulli's Theorem.

① Torricelli Theorem

$\frac{dV}{dt} \Rightarrow aV = Au$ — (i)



$P_{atm} + \frac{1}{2}\rho v^2 = P_{atm} + \rho gh + \frac{1}{2}\rho u^2$ — (ii)

$v^2 = u^2 + 2gh$ — (ii)

$v^2 = \frac{a^2v^2}{A^2} + 2gh \Rightarrow v^2\left(\frac{A^2-a^2}{A^2}\right) = 2gh$

$$\boxed{v = \frac{A\sqrt{2gh}}{\sqrt{A^2-a^2}}}$$

velocity of free surface = $\boxed{u = \frac{av}{A} = \frac{a}{\sqrt{A^2 - a^2}} \sqrt{2gh}}$

here v is known as velocity of efflux.

volume flow rate = $\frac{dV}{dt} = av = Au = \boxed{\frac{Aa}{\sqrt{A^2 - a^2}} \sqrt{2gh}}$

mass flow rate = $\frac{dm}{dt} = \rho \frac{dV}{dt} = \boxed{\frac{\rho Aa}{\sqrt{A^2 - a^2}} \sqrt{2gh}}$

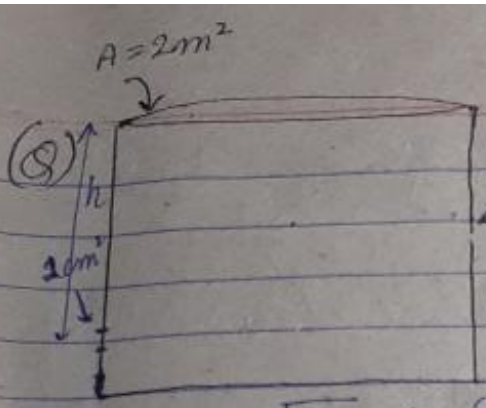
• Special case ($A >> a$)

$v = \frac{A}{\sqrt{A^2 - a^2}} \sqrt{2gh}$ now a is negligible against A

$\therefore \boxed{v = \sqrt{2gh}}$ $u = \frac{a}{\sqrt{A^2 - a^2}} \sqrt{2gh} = \frac{a}{A} \sqrt{2gh}$

volume flow rate = $\frac{dV}{dt} = Au = av = a\sqrt{2gh}$

mass flow rate = $\frac{dm}{dt} = \rho \frac{dV}{dt} = \rho a \sqrt{2gh}$

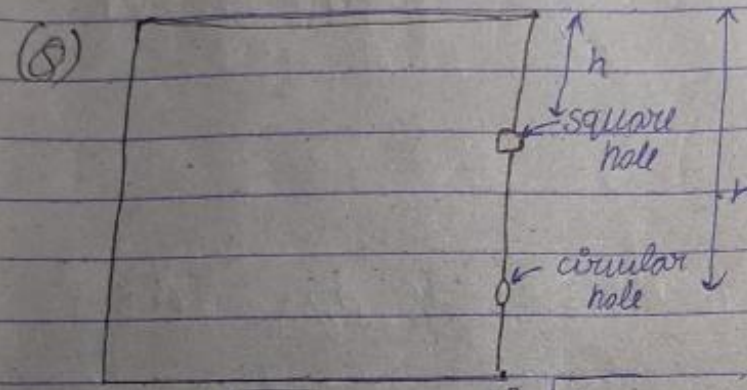


(Q) Find h such that volume flow rate is same from both the holes.

$$\frac{dv}{dt} = a\sqrt{2gh} \Rightarrow 2cm^2\sqrt{2gH} = 1cm^2\sqrt{2gh}$$

$$= 4 \times 3 = h$$

$$h = 12\ m$$

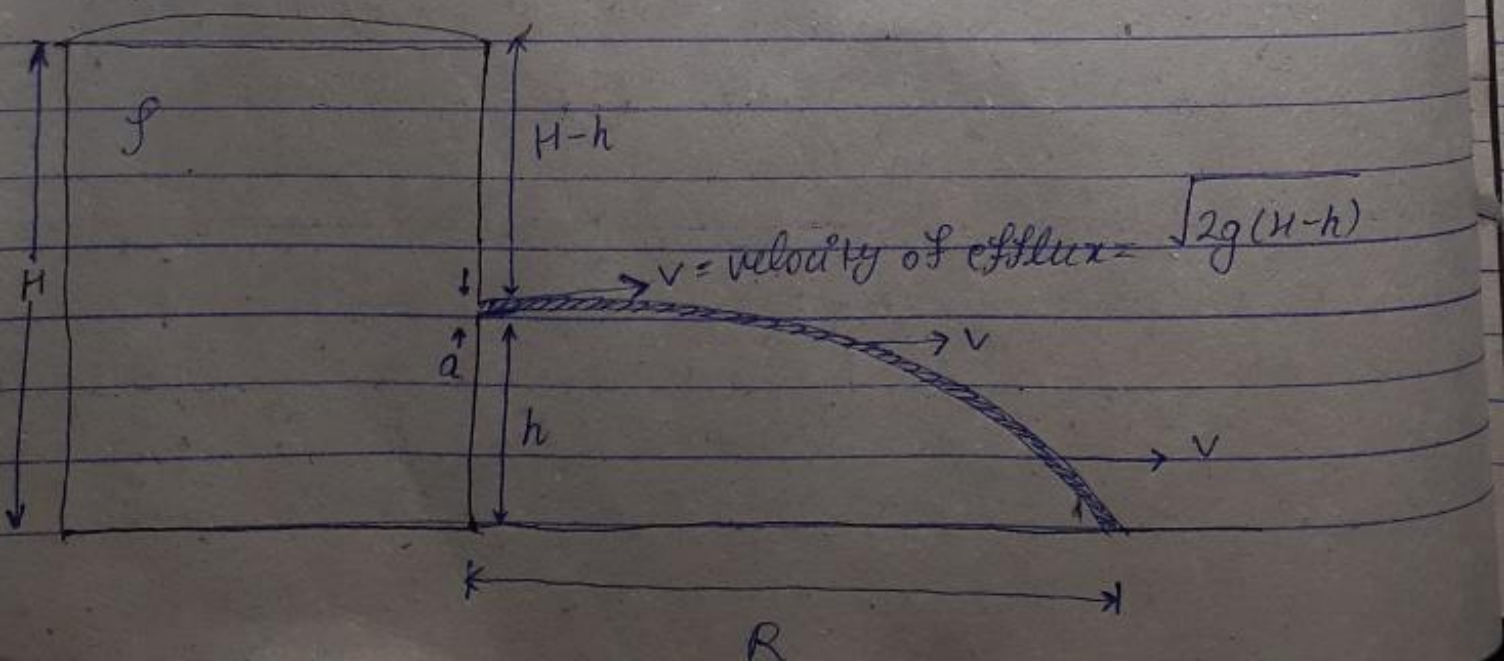


(Q) Find H such that volume flow rate through both holes are equal.

$$av = a\sqrt{2gh} \Rightarrow l^2\sqrt{2gh} = \pi r^2\sqrt{2gH}$$

$$l^4 h = \pi^2 r^4 H \Rightarrow H = \frac{l^4 h}{\pi^2 r^4}$$

- Range of water stream.



$u_y = 0$, $a_y = +g$, $S_y = +h \Rightarrow S_y = u_y t + \frac{1}{2} g t^2$

$$\boxed{t = \sqrt{\frac{2h}{g}}} \qquad S_x = u_x t = \sqrt{2(H-h)g} \times \sqrt{\frac{2H}{g}}$$

$$R = \sqrt{4(H-h)h}$$

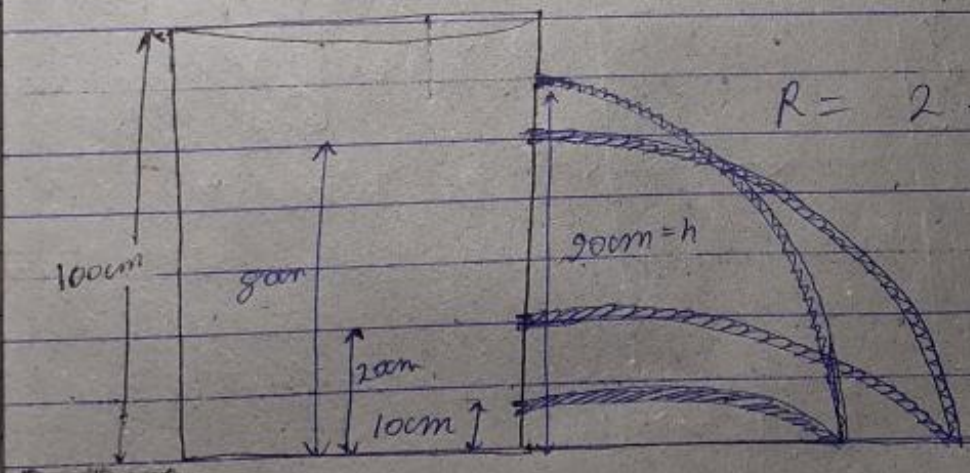
$$\boxed{R = 2\sqrt{(H-h)h}}$$

now, $R = f(h)$, $f(h) = Hh - h^2$

$f'(h) = H \times 1 - 2h = 0 \Rightarrow h = H/2$

$f''(h) = 0 - 2 = -2$

$$\therefore \boxed{R_{max} = 2\sqrt{\left(H - \frac{H}{2}\right)\frac{H}{2}} = H}$$

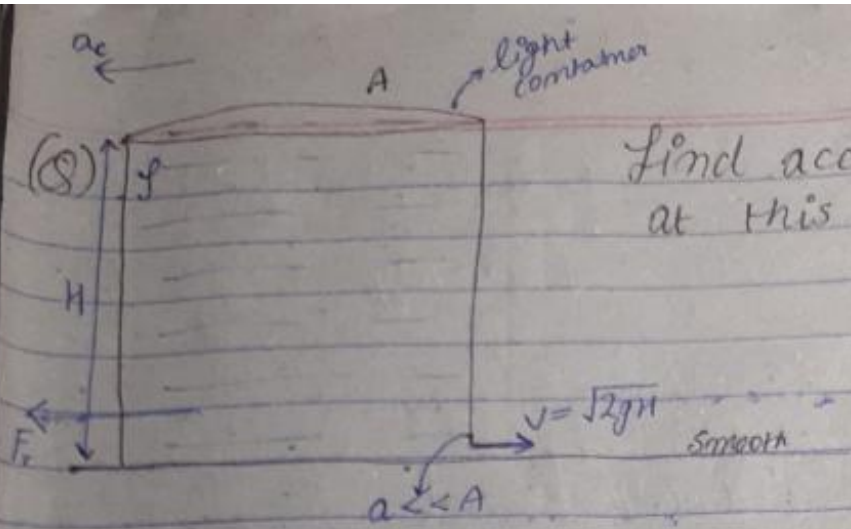


$R = 2\sqrt{90 \times 10} = 60cm$

$R_2 = 2\sqrt{10 \times 90} = 60cm$

$R_3 = 2\sqrt{80 \times 20} = 80cm$

$R_4 = 2\sqrt{20 \times 80} = 80cm$

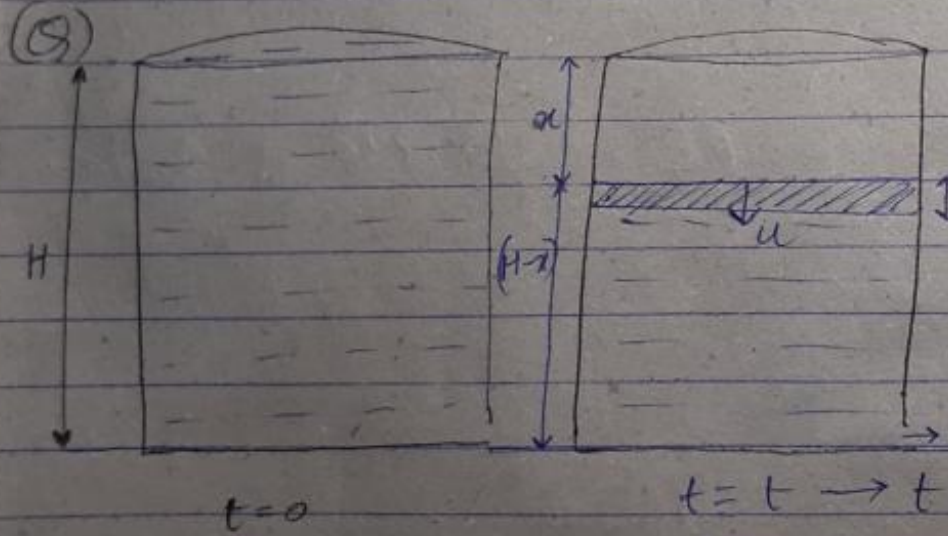


Find acc^n of container at this instant.

$$F_T = \left(\frac{dm}{dt}\right) v = \rho\left(\frac{dV}{dt}\right) v = \rho a v^2 = \rho a 2gH$$

$$F_T = m a_c \Rightarrow 2\rho g H a = (\rho A H)\, a_c$$

$$\therefore \boxed{a_c = \frac{a}{A} 2g}$$



(i) Find ① v ② u at time t

$$v = \sqrt{2g(H-x)}, \quad av = Au \Rightarrow u = \frac{a}{A}\sqrt{2g(H-x)}$$

(ii) Find volume flow rate at t

$$\frac{dV}{dt} = av = a\sqrt{2g(H-x)}$$

(iii) Find time in which container will get empty

$dV = A\,dx$, $\frac{A\,dx}{dt} = a\sqrt{2g(H-x)}$

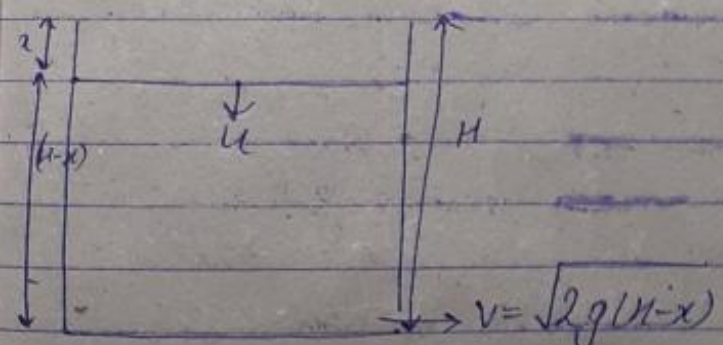
$$A\int_0^H dx\,(H-x)^{-1/2} = \frac{a}{A}\sqrt{2g}\int_0^t dt$$

$$-A\left[(H-x)^{1/2}\right]_0^H \times 2 = a\sqrt{2g}\,t$$

$$-\frac{2A}{a}\left\{0-\sqrt{H}\right\} = \sqrt{2g}\,t.$$

$$t = \sqrt{\frac{2H}{g}}\cdot\frac{A}{a}$$

Alter:



$$u = \frac{a}{A}\sqrt{2g(H-x)}$$

$t=0 \Rightarrow x=0$; $u = \frac{a}{A}\sqrt{2gH}$

$t=t$, $x=h$, $u=0$

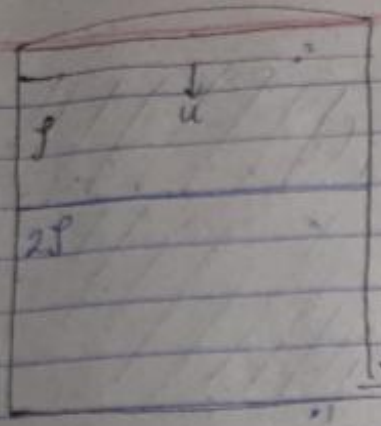
Find acc of free surface

$$u^2 = \frac{a^2}{A^2}2g(H-x) \Rightarrow 2u\frac{du}{dx} = \frac{a^2}{A^2}2g\,(0-1)$$

$$u\frac{du}{dx} = a = -\frac{a^2}{A^2}g$$

$$v = u + at \Rightarrow 0 = \frac{a}{A}\sqrt{2gH} - \frac{a^2}{A^2}g\,t \Rightarrow t = \frac{A}{a}\sqrt{\frac{2H}{g}}$$

(Q)



$Au = av$

as $A >> a$

$\therefore u << v$

$P_{atm} + 0 + \frac{1}{2} \times 2\rho \times v^2 = P_{atm} + (2\rho gh + \rho gh) + \frac{1}{2}\rho u^2$ (→ 0)

$\rho v^2 = 3\rho gh \Rightarrow v = \sqrt{3gh}$

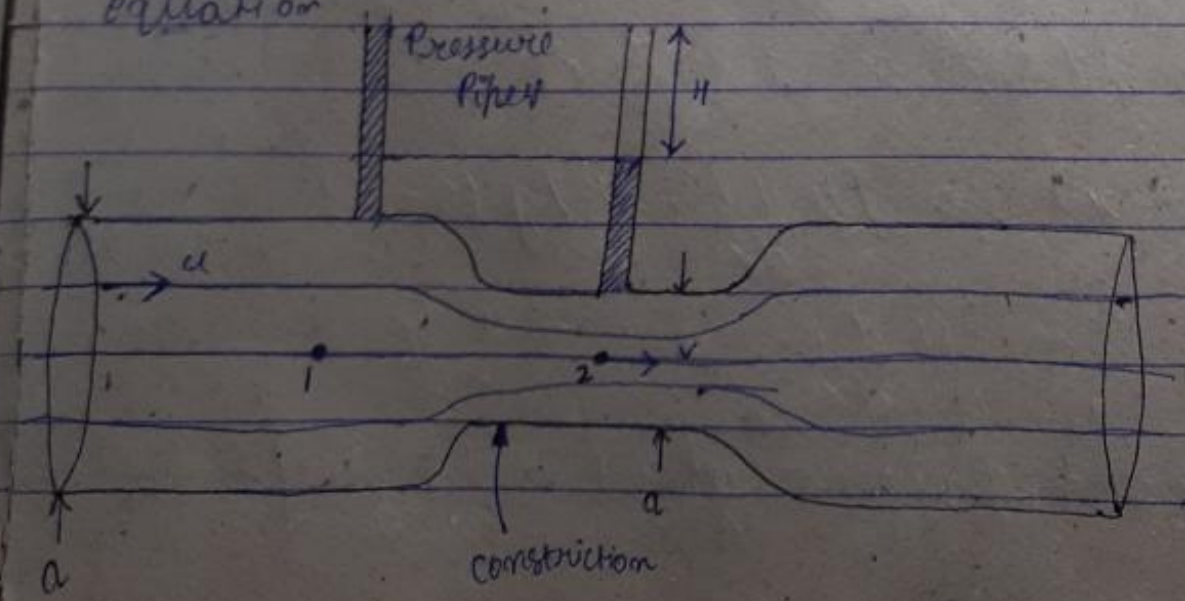
Alter: $\rho gh = (2\rho) g h'$

$h' = \frac{h}{2}$ ⇒ replace the upper liquid by 2ρ liq at height $\frac{h}{2}$

$\therefore v = \sqrt{2 \cdot g \cdot \frac{3h}{2}} = \sqrt{3gh}$

- **Venturimeter**

Arrangement used to find volume flow rate through a pipe. It is based on bernoulli's equation



$\Rightarrow Au = av$ ——— (i)

$P_1 + \rho g h' + \frac{1}{2}\rho u^2 = P_2 + \rho g h' + \frac{1}{2}\rho v^2$

$P_1 - P_2 = \frac{1}{2}\rho(v^2 - u^2)$ ——— (ii)

$P_1 - P_2 = \rho g H$ ——— (iii) ; H is difference of height upto which liq has been filled in pipe.

$\rho g H = \frac{1}{2}\rho(v^2 - u^2)$

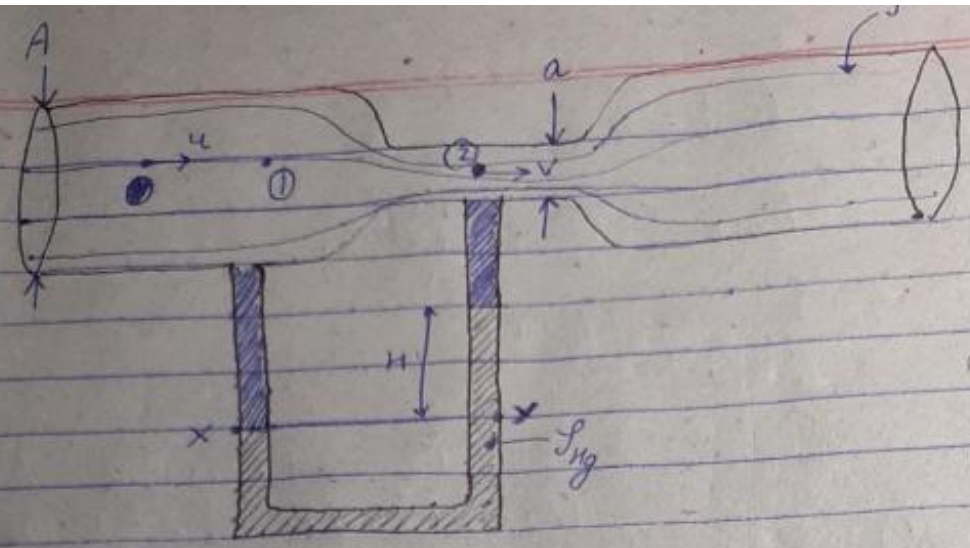
$v^2 - u^2 = 2gH$

$\Rightarrow \frac{A^2u^2}{a^2} - u^2 = 2gH$

$\left(\frac{A^2 - a^2}{a^2}\right) u^2 = 2gH \Rightarrow u = \frac{a}{\sqrt{A^2 - a^2}} \cdot \sqrt{2gH}$

volume flow rate $= \frac{dV}{dt} = Au = av = \frac{Aa}{\sqrt{A^2 - a^2}} \cdot \sqrt{2gH}$

mass flow rate $= \frac{dm}{dt} = \frac{\rho dV}{dt} = \frac{\rho Aa}{\sqrt{A^2 - a^2}} \sqrt{2gH}$



$$Au = av \quad \text{——} (1)$$

$$P_1 + \rho g h' + \frac{1}{2}\rho u^2 = P_2 + \rho g h' + \frac{1}{2}\rho v^2$$

$$P_1 - P_2 = \frac{1}{2}\rho(v^2 - u^2) \quad \text{——} (2)$$

$$P_x = P_y \Rightarrow P_1 + \rho g h_1 = P_2 + \rho g (h_1 - h) + \rho_{Hg} g H$$

$$\Rightarrow P_1 - P_2 = (\rho_{Hg} - \rho) g H \quad \text{——} (3)$$

$$\frac{1}{2}\rho(v^2 - u^2) = (\rho_{Hg} - \rho) g H$$

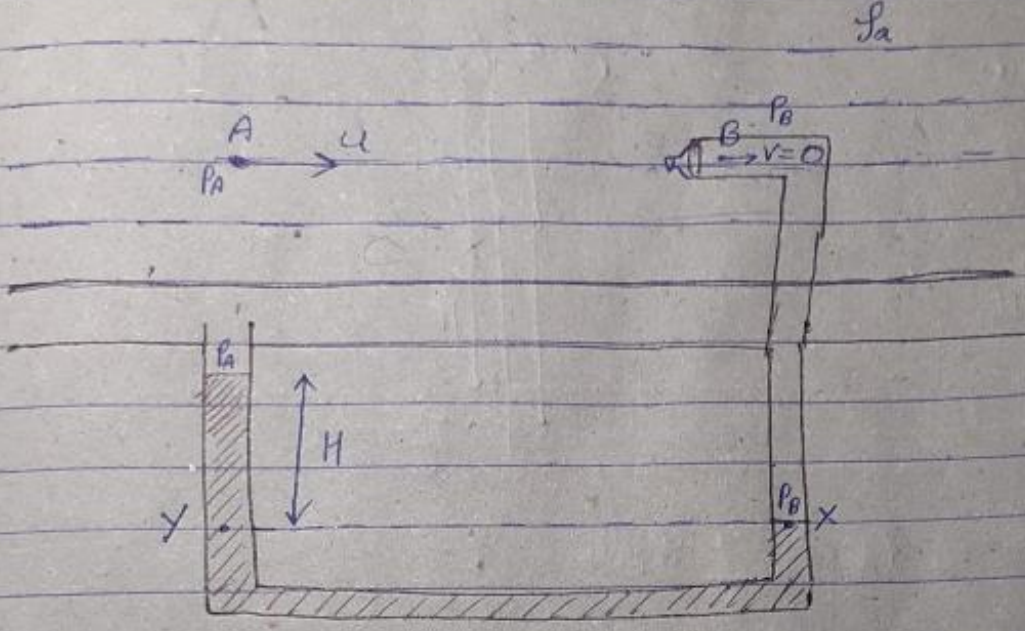
$$\rho\left(\frac{A^2u^2}{a^2} - u^2\right) = 2(\rho_{Hg} - \rho) g H$$

$$u^2\left(\frac{A^2 - a^2}{a^2}\right) = 2\left(\frac{\rho_{Hg}}{\rho} - 1\right) g H$$

$$\boxed{u = \frac{a}{\sqrt{A^2 - a^2}} \cdot \sqrt{2\left(\frac{\rho_{Hg}}{\rho} - 1\right) g H}}$$

Pitot Tube

It is used to find air speed.



$$P_A + \rho g h' + \frac{1}{2}\rho u^2 = P_B + \frac{1}{2}\rho 0^2 + \rho g h'$$

$$P_B - P_A = \frac{1}{2}\rho_a u^2 \quad \text{——— (1)}$$

$$P_x = P_y \Rightarrow P_B = P_A + \rho_{Hg} g H$$

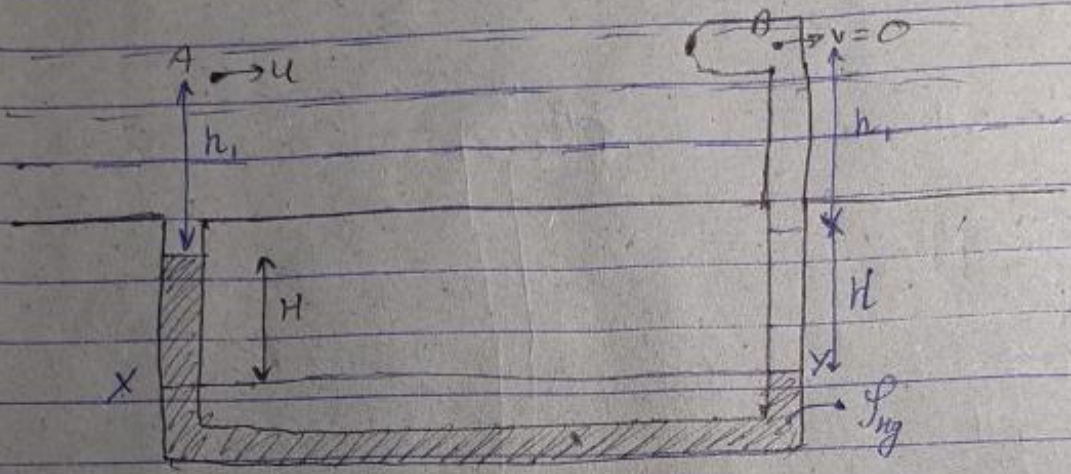
$$P_B - P_A = \rho_{Hg} g H \quad \text{——— (2)}$$

$$\frac{1}{2}\rho_a u^2 = \rho_{Hg} g H$$

$$\therefore u = \sqrt{\frac{2\rho_{Hg} g H}{\rho_a}}, \text{ when } \rho_a << \rho_{Hg}$$

if ρ_a is comparable to ρ_{Hg}

ρ_{air}



$$P_A + \frac{1}{2}\rho u^2 = P_B + \frac{1}{2}\rho 0^2$$

$$P_B - P_A = \frac{1}{2}\rho_{air} u^2 \quad \text{——— (1)}$$

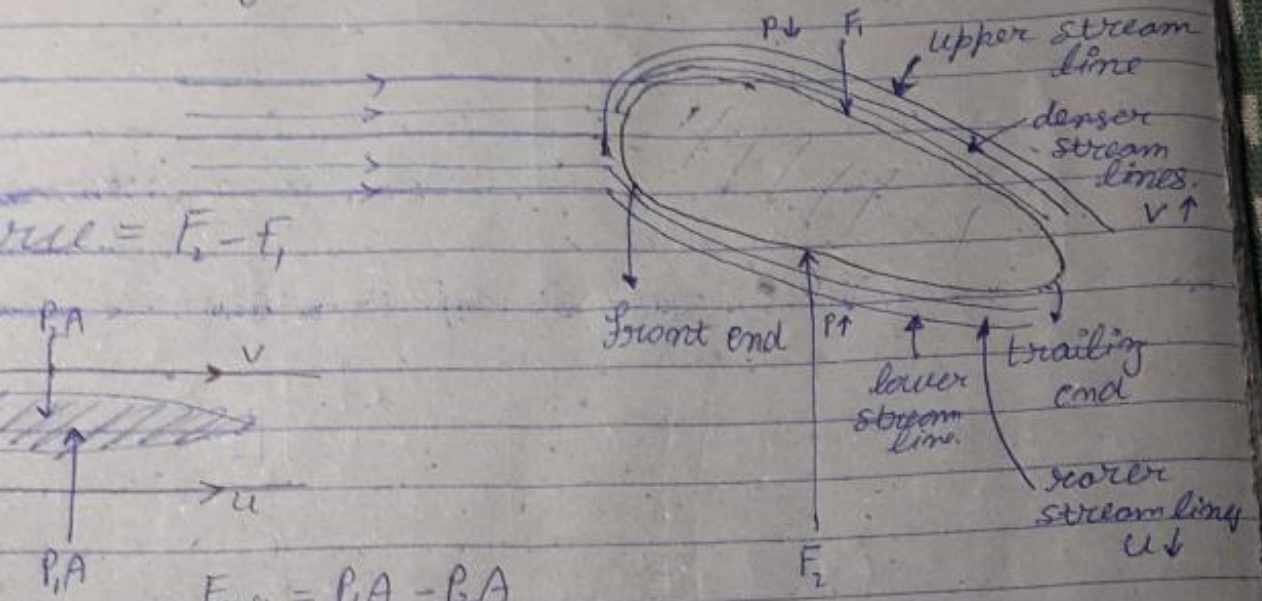
$$P_x = P_y \Rightarrow P_A + \rho_{air} g h_1 + \rho_{Hg} g H = P_B + \rho_{air} g (h_1 + H)$$

$$P_B - P_A = (\rho_{Hg} - \rho_{air}) g H \quad \text{——— (2)}$$

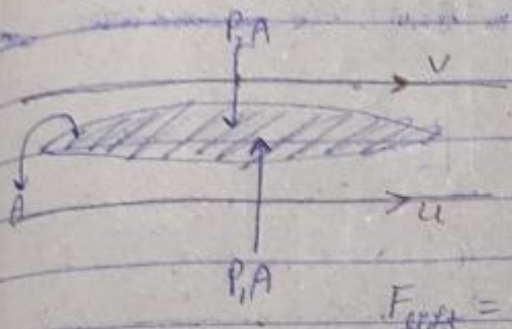
$$\frac{1}{2}\rho_{air} u^2 = (\rho_{Hg} - \rho_{air}) g H$$

$$u = \sqrt{2\left(\frac{\rho_{Hg} - \rho_{air}}{\rho_{air}}\right) g H}$$

• Force on wings of airplane (lift force)



Lift force = $F_2 - F_1$

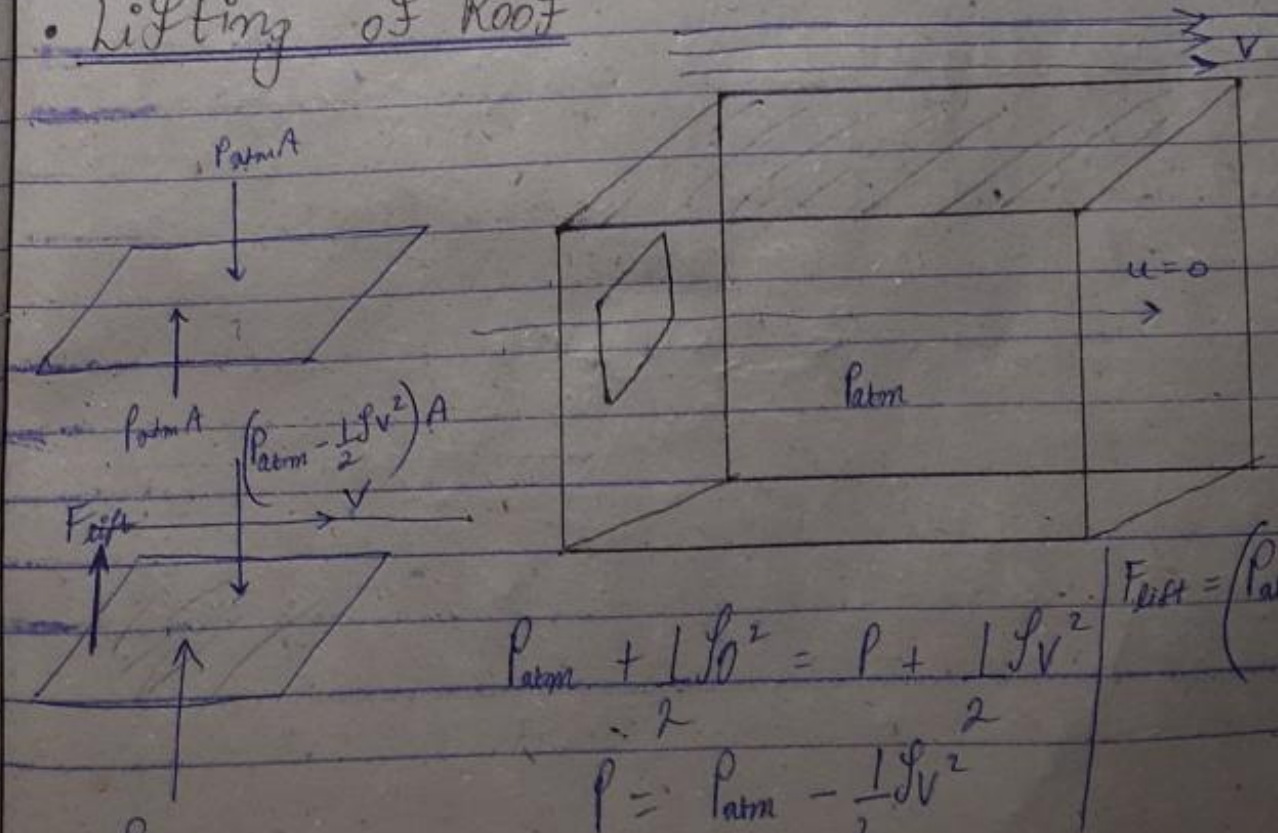


$F_{lift} = P_1A - P_2A$

$$P_2 + \frac{1}{2}\rho v^2 = P_1 + \frac{1}{2}\rho u^2$$

$$P_1 - P_2 = \frac{1}{2}\rho(v^2 - u^2) \Rightarrow (P_1 - P_2)A = \frac{1}{2}\rho(v^2 - u^2)A$$

• Lifting of Roof

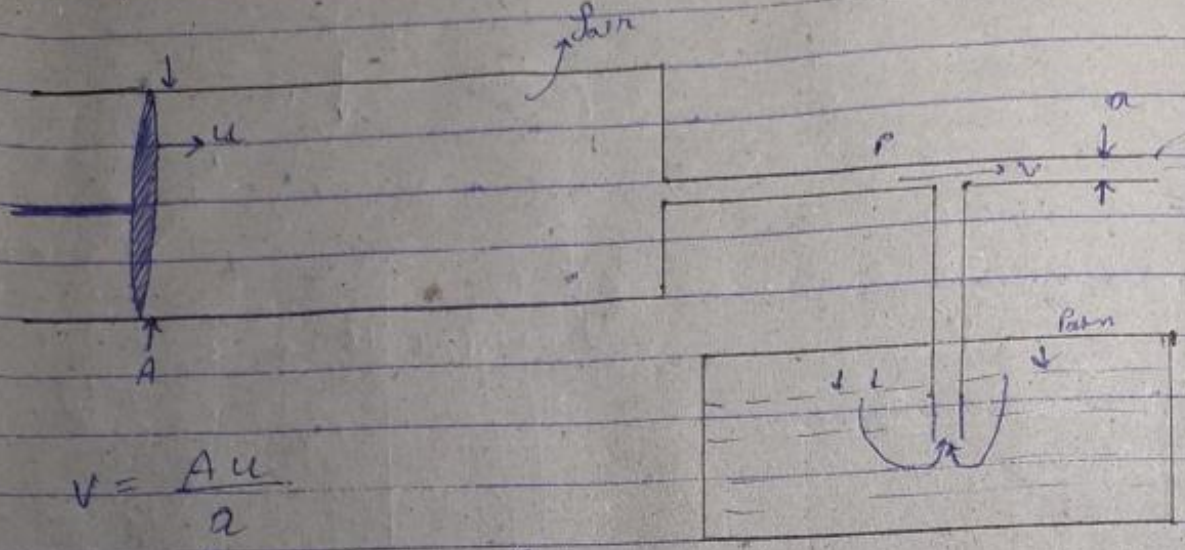


$$P_{atm} + \frac{1}{2}\rho 0^2 = P + \frac{1}{2}\rho v^2$$

$$P = P_{atm} - \frac{1}{2}\rho v^2$$

$$F_{lift} = \left(P_{atm} - \frac{1}{2}\rho v^2\right)A$$

- **Aspirator Pump / Atomizer**



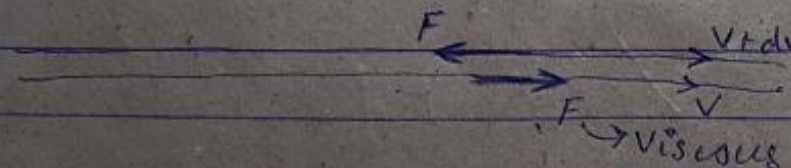
$$v = \frac{Au}{a}$$

$$P_{atm} + \frac{1}{2}\rho v^2 = P + \frac{1}{2}\rho v^2$$

$$P = P_{atm} - \frac{1}{2}\rho_{air} v^2 = P_{atm} - \frac{1}{2}\rho_{air}\left(\frac{A^2u^2}{a^2}\right)$$

- **Viscosity**

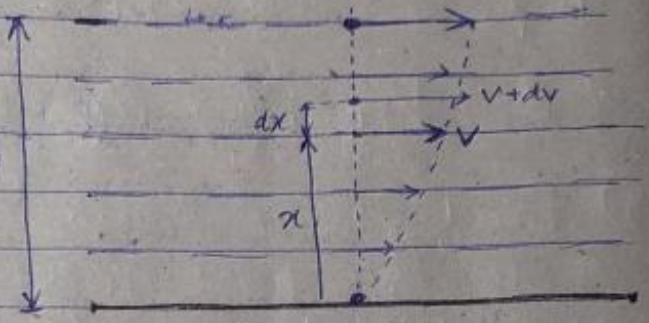
It is the property of fluid due to which fluids apply a resistive, non conservative force when there is relative motion between layers of fluids in contact.



It is a shear / tangential force which acts along tangential direction and is always opposite to relative velocity.

In liquids, density ∝ viscocity
In gases, density ∝ 1/viscocity

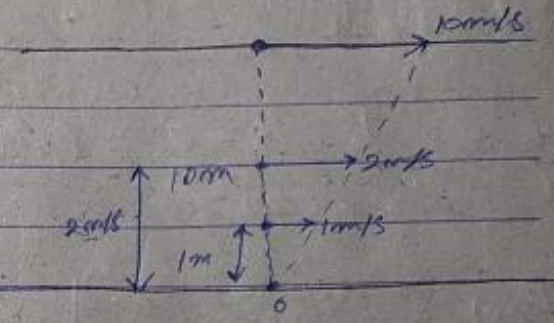
• Velocity Gradient



$\frac{dv}{dx}$ = velocity gradient

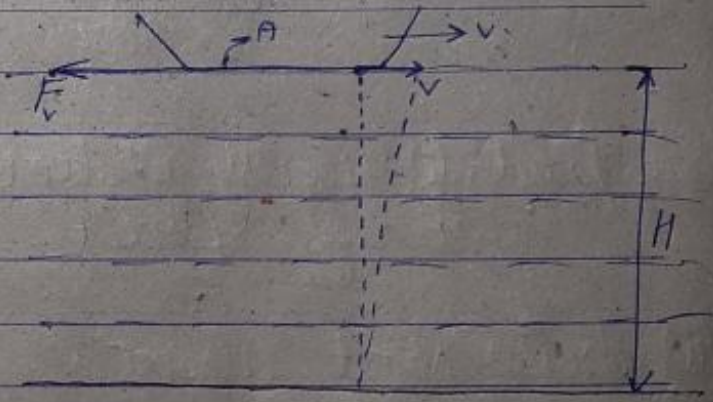
change in velocity wrt position. It is uniform

unit : s^{-1} , dim : $[T^{-1}]$



$\frac{dv}{dx} = \frac{10m/s - 0}{10m} = 1\ sec^{-1}$

• Force of viscocity



$F_v \propto A$

$\propto \frac{dv}{dx}$

$\therefore F_v = \eta A \frac{dv}{dx}$

η is known as coefficient of viscocity which depends on type of liquid.

For liquids, density ∝ η ; unit : $N-s/m^2$

dim : $[ML^{-1}T^{-1}]$

CGS unit : $\frac{10^5 \text{ dyne-s}}{10^4 \text{ cm}^2} = 10 \text{ dyne-s/cm}^2$

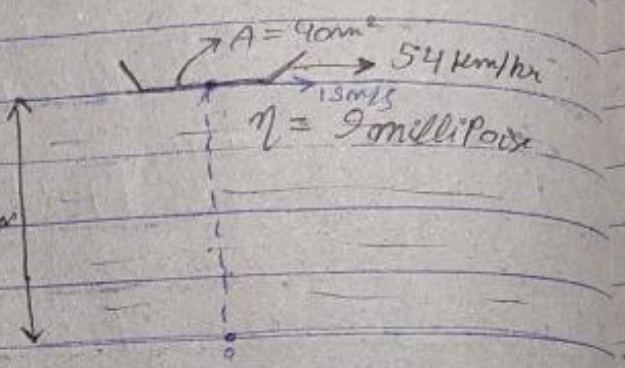
$= 10$ poise

$1 \text{ N-s/m}^2 = 1$ decapoise

10 poise $= 1 \text{ N-s/m}^2$

$\eta \propto \frac{1}{\sqrt{T}}$, for liquids and $\eta \propto \sqrt{T}$ for gases.

(Q) Find force required to move this boat with constant velocity of 54 km/hr.



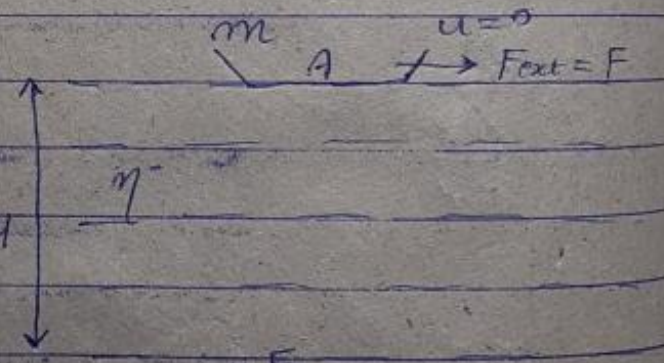
$\frac{dv}{dx} = \frac{15 \text{ m/s} - 0}{2 \text{ cm}} = 0.75 \text{ s}^{-1}$

$F_{ext} = F_v = \eta A \frac{dv}{dx} = 9 \times 10^{-4} \frac{\text{N-s}}{\text{m}^2} \times 40 \text{ m}^2 \times \frac{0.75}{\text{s}}$

(Q) Find; (i) velocity of boat at any time t.

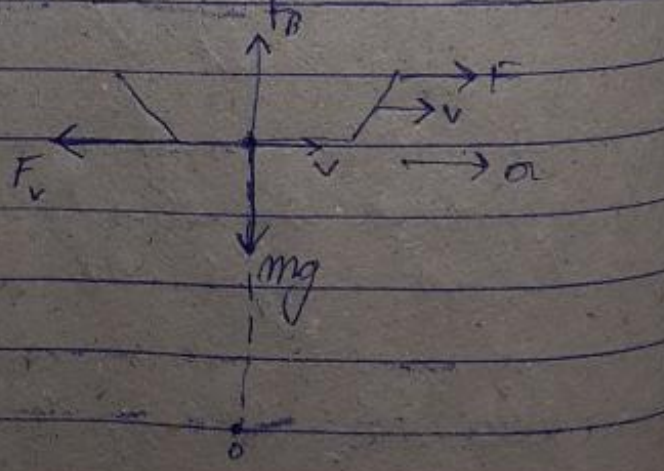
(ii) draw v-t curve

(iii) max v of boat.



$\frac{dv}{dx} = \frac{v}{H}$

$F_v = \eta A \frac{dv}{dx} = \frac{\eta A v}{H}$



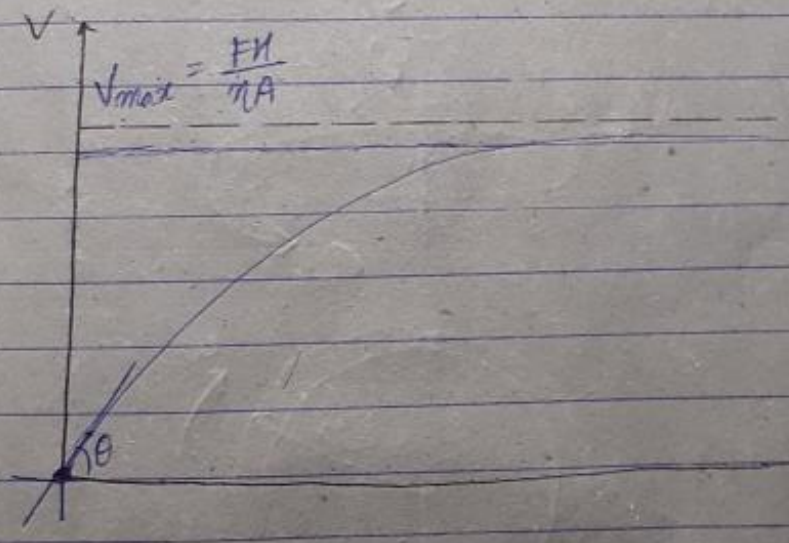
$$F = \frac{\eta A v}{H} = m\frac{dv}{dt}$$

$$\int_0^t dt = m\int_0^v \frac{dv}{F - \frac{\eta A v}{H}} \Rightarrow t = \frac{mH}{-\eta A}\ln\left[F - \frac{\eta A v}{H}\right]_0^v$$

$$\Rightarrow \frac{-\eta A t}{mH} = \ln\left(F - \frac{\eta A v}{H}\right) - \ln F$$

$$\frac{-\eta A t}{mH} = \ln\left(1 - \frac{\eta A v}{FH}\right) \Rightarrow 1 - \frac{\eta A v}{FH} = e^{\frac{-\eta A t}{mH}}$$

$$v = \frac{FH}{\eta A}\left(1 - e^{-\frac{\eta A t}{mH}}\right)$$



when $t \longrightarrow \infty$

$e^{-\infty} = 1/e^{\infty} = 0$

$\tan\theta = \frac{F}{m}$

$t \to \infty$ v_{max}, $F_v = F$

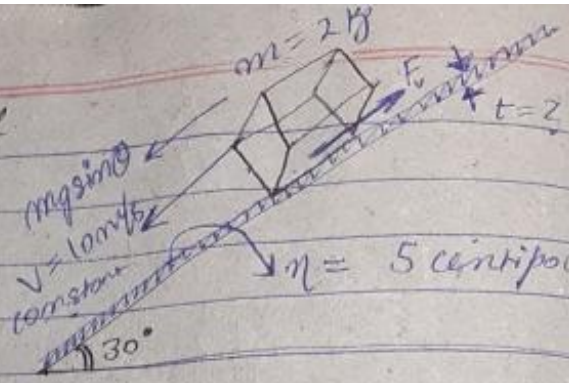
$F_{net} = 0$, $a = 0$

$F = \frac{\eta A v}{H}$

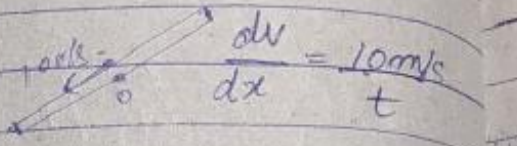
Power of $F_v = \vec{F_v} \cdot \vec{v} = \frac{\eta A v}{H} \times v \times \cos\pi$

$= -\frac{\eta A v^2}{H}$

(8) Find the thickness of oil layer.
side of cube = 2m



$$F_v = \frac{\eta A dv}{dx} = mg\sin\theta$$



$$F_v = 5\times10^{-3}\ \frac{N\text{-}s}{m^2} \times 4\,m^2 \times \frac{10\,m/s}{t}$$

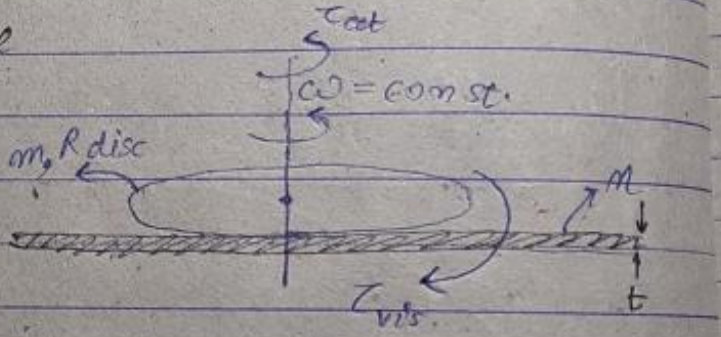
$$= 10\ N$$

$$\Rightarrow 5\times10^{-3}\times 4\times 10\,m = t$$

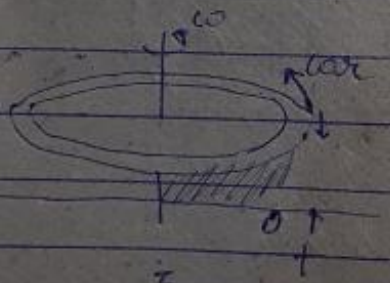
$$t = 200\times10^{-3}\,m = 200\,mm = 20\,cm$$

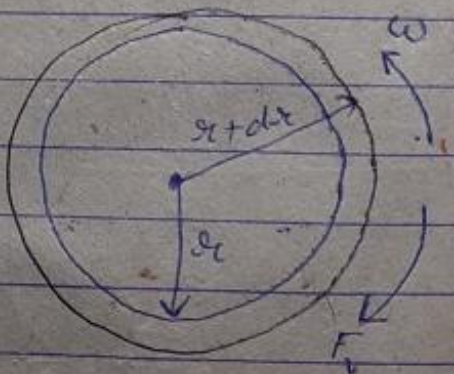
$$t = 20\times10^{-3}\,m = 20\,mm$$

(8) Find external torque required to rotate the disc with cont ang. velocity ω



$$d\tau_v = \eta\, 2\pi r\, dr\, \frac{(\omega r)}{t}\cdot r$$

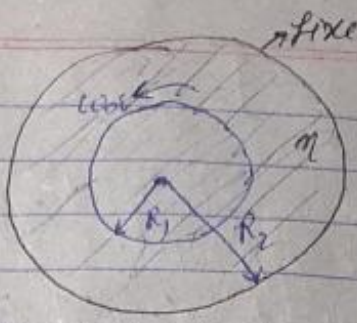


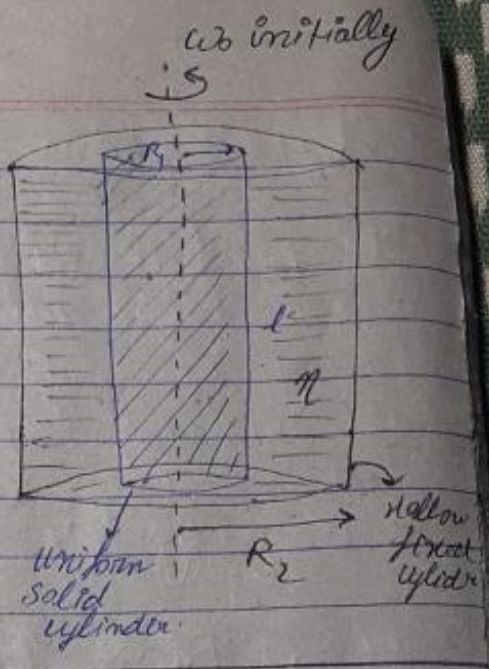


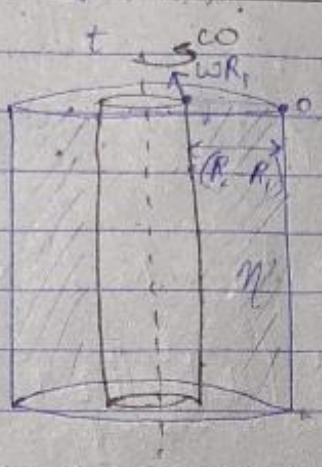
$$\tau_v = \int_0^z d\tau_v = \frac{\eta 2\pi\omega}{t}\int_0^R r^3\, dr$$

$$\tau_v = \frac{\eta\pi\omega R^4}{2t} = \tau_{ext}$$

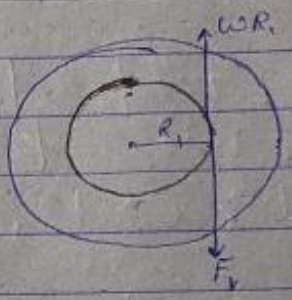
(8) Find angular velocity of solid cylinder at any time t.







$$\frac{dv}{dx} = \frac{\omega R_1}{R_2 - R_1}$$

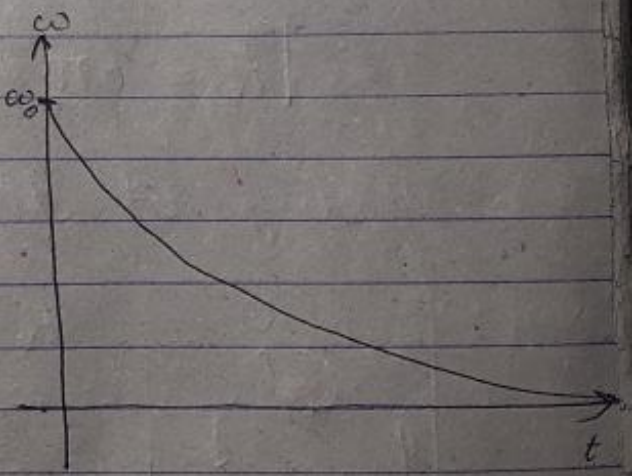


$$\tau = \left(\frac{\eta(2\pi R_1 l)\omega R_1}{R_2 - R_1}\right) R_1 = m\frac{R_1^2}{2}\alpha$$

$$\alpha = \frac{2\eta\pi R_1 l\omega}{m(R_2 - R_1)} = -\frac{d\omega}{dt}$$

$$-\frac{4\pi R_1 \eta l}{m(R_2 - R_1)} \int_0^t dt = \int_{\omega_0}^{\omega} \frac{d\omega}{\omega}$$

$$\omega = \omega_0 e^{\frac{-4\pi R_1 l \eta}{m(R_2 - R_1)}}$$



• Stoke's Law

It is for solids completely dipped inside liquid. Viscous force will be opposite to direction of relative velocity of solid wrt fluid. Work done by this viscous force will dissipate in form of heat.

It is only for spherical solid

now, $F_v \propto \eta^a r^b v^c \Rightarrow F_v = k \eta^a r^b v^c$

$\therefore [MLT^{-2}] = [ML^{-1}T^{-1}]^a [L]^b [LT^{-1}]^c$

$[MLT^{-2}] = [M^a L^{-a+b+c} T^{-a-c}]$

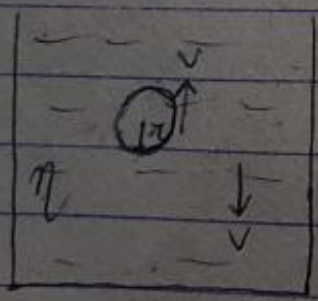
$a = 1, \quad c = 1, \quad b = 1$

$\therefore F = k \eta r v$, experimentally $k = 6\pi$

$\therefore \boxed{F_v = 6\pi \eta r v}$, where r is radius of sphere.

v is velocity of sphere and η is coff. of viscosity

NOTE:



$F_v = 6\pi \eta r v_{rel}$

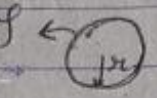
$\therefore$ wrt fluid v_R of sphere $= 2v$

$\therefore F_v = 12\pi \eta r v$

Terminal velocity of a ball in air

density of air is almost negligible.

mass of ball = $m = \rho \frac{4}{3}\pi r^3$

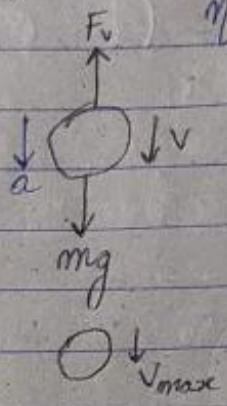


at t=0 released.

η of air.

at $t=0$; $u=0 \Rightarrow F_v = 0$,

$F_{net} = mg$, $a = g$.



at $t = t'$

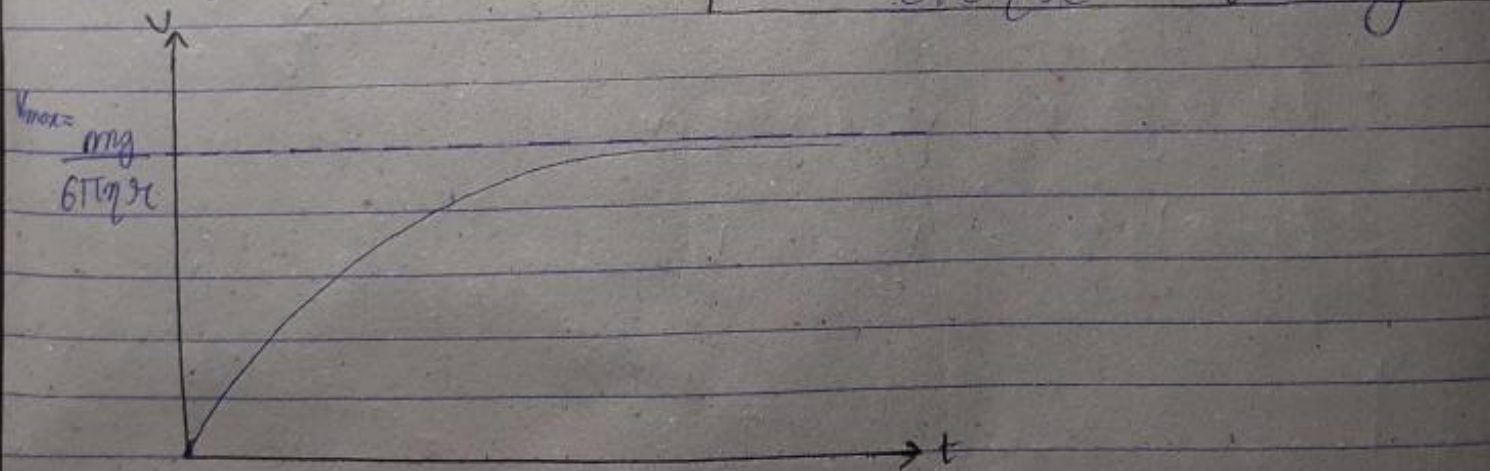
at $t = t$; $F_v = 6\pi\eta r v$

$t \uparrow \Rightarrow v \uparrow \Rightarrow F_v \uparrow \Rightarrow F_{net} \downarrow$

$\Rightarrow a \downarrow$

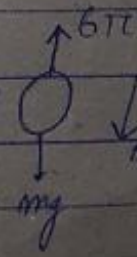
at $t \longrightarrow \infty$; $F_v = mg \Rightarrow F_{net} = 0, a = 0 \Rightarrow$ const velocity

$\therefore mg = 6\pi\eta r V_{max} \Rightarrow \boxed{V_{max} = \frac{mg}{6\pi\eta r}}$ = terminal velocity



$$mg - 6\pi\eta r v = m\frac{dv}{dt}$$

$$\int_0^t dt = m\int_0^v \frac{dv}{mg - 6\pi\eta r v}$$



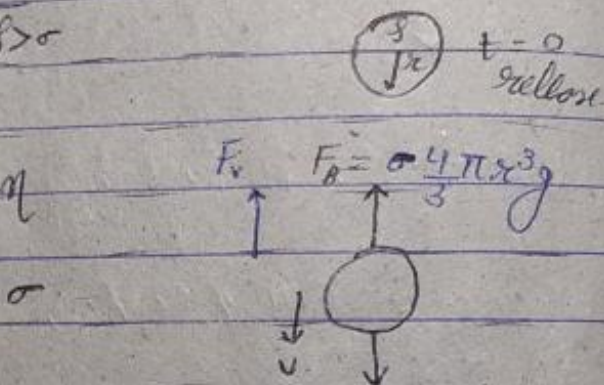
$$|t|_0^t = \frac{m}{-6\pi\eta r} \cdot \ln|mg - 6\pi\eta r v|_0^v$$

$$\frac{-6\pi\eta r t}{m} = \ln \frac{mg - 6\pi\eta r v}{mg}$$

$$\boxed{v = \frac{mg}{6\pi\eta r}\left(1 - e^{\frac{-6\pi\eta r t}{m}}\right)}$$

- **Terminal velocity of ball in liquid**

$t=0$; $F_{net} = mg - F_B$ $\Rightarrow \rho > \sigma$



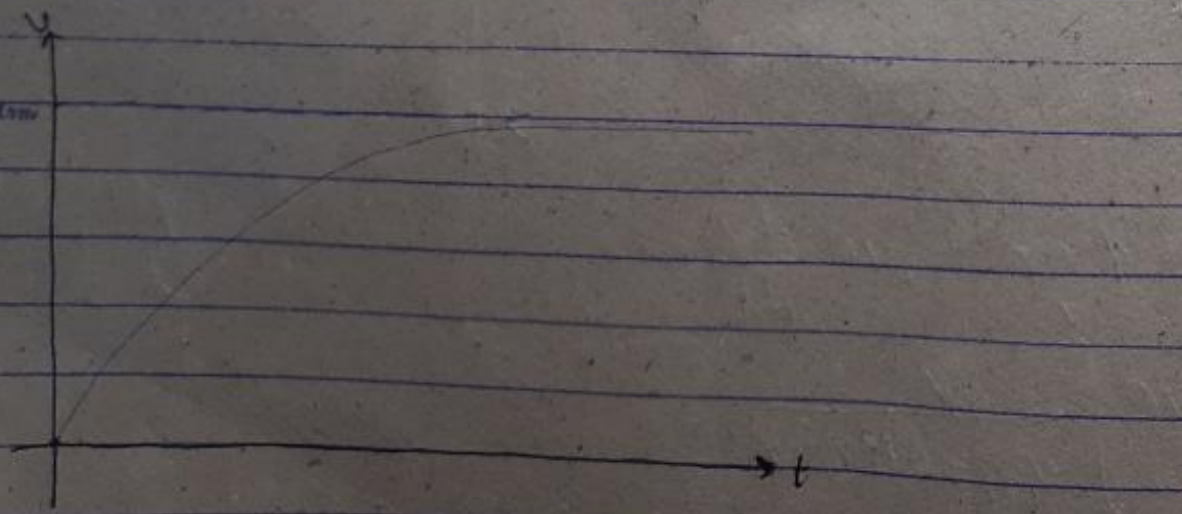
$t \uparrow \Rightarrow v \uparrow \Rightarrow F_{net} = mg - F_B - F_v$

$t \to \infty \Rightarrow v =$ constant, $F_{net} = 0$

$mg = F_B + F_v$

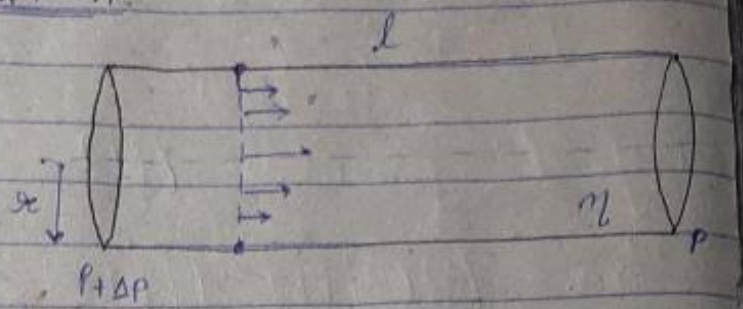
$$\rho \frac{4}{3}\pi r^3 g = \sigma \frac{4}{3}\pi r^3 g + 6\pi\eta r v$$

$$\frac{2(\rho - \sigma) r^2 g}{9\eta} = v_{max} = v_{terminal}$$



Power of F_v: $F_v \, v \cos\pi = -6\pi\eta r v^2$

Hagen - Poiseuille's equation



Volume flow rate = $\frac{\Delta V}{\Delta t}$

Pressure gradient = $\frac{\Delta P}{l}$

$$\frac{\Delta V}{\Delta t} = k \eta^a r^b \left(\frac{\Delta P}{l}\right)^c$$

$$\Rightarrow [M^0 L^3 T^{-1}] = [ML^{-1}T^{-1}]^a [L]^b [ML^{-2}T^{-2}]^c$$

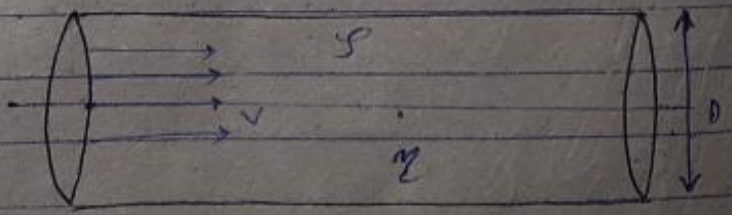
$$[M^0 L^3 T^{-1}] = [M^{a+c} L^{-a+b-2c} T^{-a-2c}]$$

$a + c = 0$, $a + 2c = 1$; $c = 1$, $a = -1$, $b = 4$

$$\therefore \frac{\Delta V}{\Delta t} = k \eta^{-1} r^4 \left(\frac{\Delta P}{l}\right)^1 = k \frac{\Delta P r^4}{l \eta}$$

experimentally $k = \frac{\pi}{8}$, $\therefore \boxed{\frac{\Delta V}{\Delta t} = \frac{\pi \Delta P r^4}{8 l \eta}}$

• Critical velocity / Reynold number



For small velocities fluid will have streamline flow. The max. velocity at which motion of fluid is steady is known as critical velocity (V_c).

$V \leq V_c \Rightarrow$ steady flow, streamline flow.

$V > V_c \Rightarrow$ flow of fluid will be turbulent.

Flow of fluid to be steady or turbulent depends on ρ, D, η of fluid.

Reynold number (N) = $\frac{\rho v D}{\eta}$

dim. = $\frac{[ML^{-3}][LT^{-1}][L]}{[ML^{-1}T^{-1}]} = [M^0L^0T^0]$

If $N < 2000 \Rightarrow$ steady flow.
$N > 3000 \Rightarrow$ turbulant.

- **Surface Tension.**

free surface will behave slightly different from bulk of fluid. This difference is due to phenomenom of surface tension.

Surface of fluid wants to have minimum surface area and behaves like stretched membrane. This phenomenon is called surface tension.

- **Force of surface Tension.**

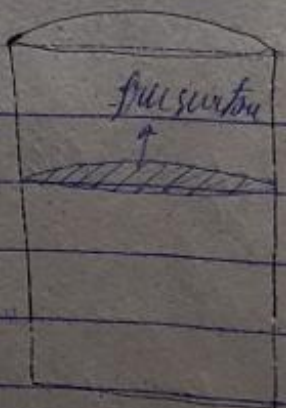
$F \propto l$

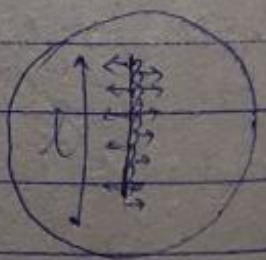
$\therefore F = Sl$

where S is coefficient of surface tension.

unit : N/m

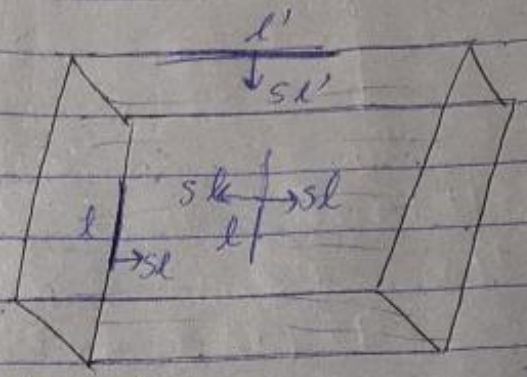
dim : $[MT^{-2}]$





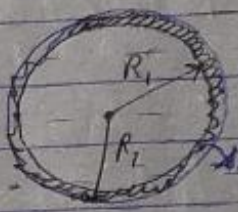
$S_{water} = 0.075$ N/m , $S_{soap} = 0.03$ N/m

Surface tension is ~~direc~~ inversely prop. to temperature
If an impurity (insoluble) is mixed in fluid then surface tension decreases.

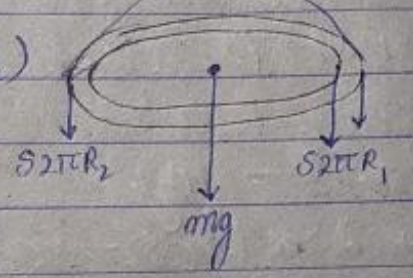


Top

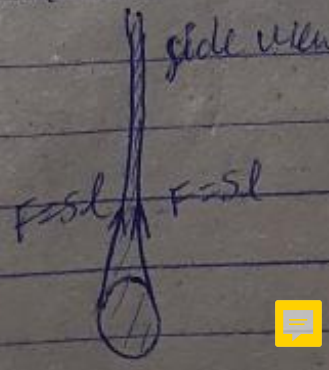
(Q) Find minimum force required to pull the ring

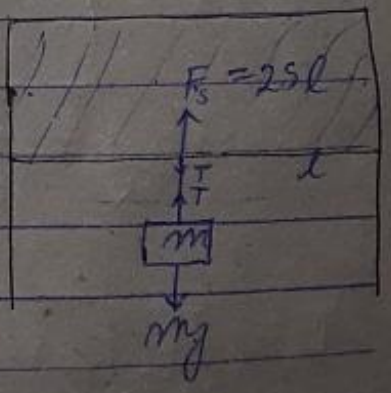


$F_{min} = mg + 2\pi S(R_1 + R_2)$



(Q) A frame of wire is dipped in soap solution and when taken out it is found that a thin film of soap solution, surface tension S is found. A block is hanged on the wire. Find m such ~~that~~ that system remains in equilibrium.

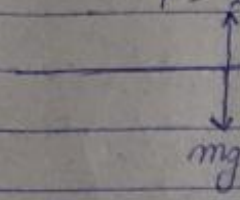


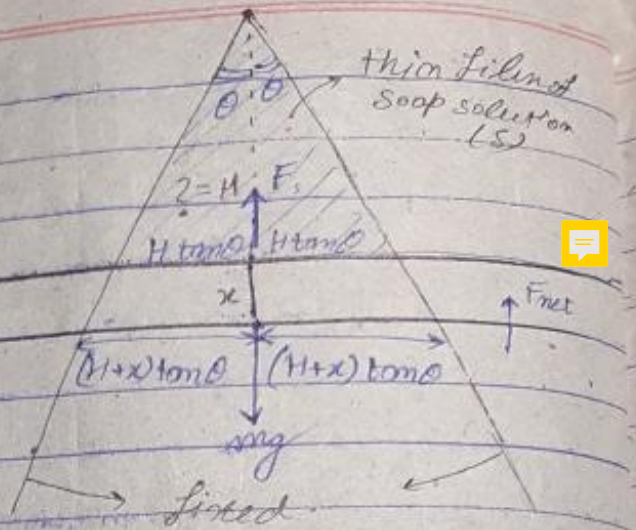


$2Sl = mg$

$m = \frac{2Sl}{g}$

(a) If horizontal wire is at equilibrium. Find H





$\therefore 4SH\tan\theta = mg$

$H = \dfrac{mg}{4S\tan\theta}$

② If horizontal wire is displaced slightly in vertical direction. Prove that it will perform SHM and find time period of this SHM.

$F_{net} = F_s - mg \Rightarrow$ SHM Proved.

$F_{Restoring} = 2\times 2S(H+x)\tan\theta - mg$

$= (2SH\tan\theta + 2Sx\tan\theta)^2 - mg$

$= 4SH\tan\theta + 4Sx\tan\theta - mg.$

$F_R = (4S\tan\theta)x \Rightarrow$ SHM Proved.

$\therefore (4S\tan\theta)x = ma$

$a = \left(\dfrac{4S\tan\theta}{m}\right)x$

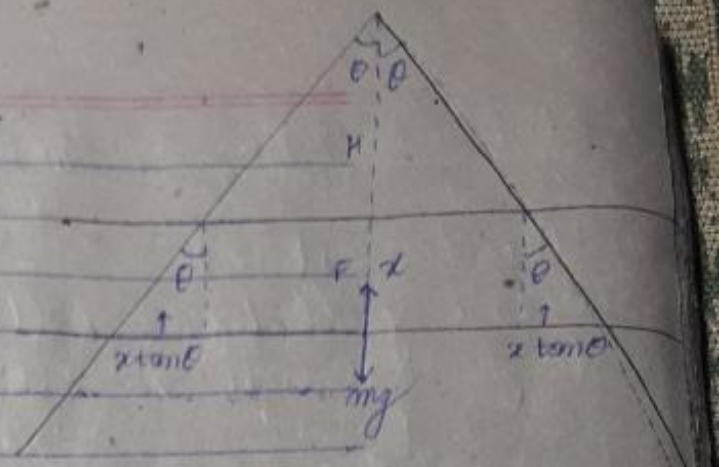
$\omega = \sqrt{\dfrac{4S\tan\theta}{m}} \Rightarrow T = \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{m}{4S\tan\theta}}$

$T = \dfrac{\pi}{2}\sqrt{\dfrac{m}{S\tan\theta}}$

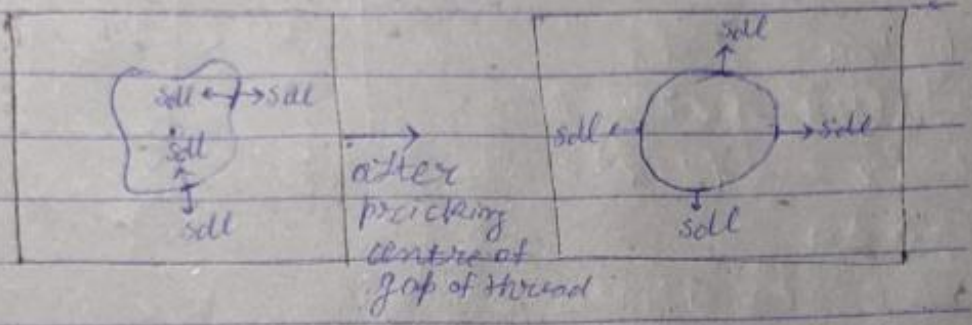
Note: $F_R = 2S(2x \tan\theta)$

$F_R = 4Sx\tan\theta$

$\Rightarrow$ SHM Proved



NOTE:



For given perimeter of thread, $(A)_{max}$ is of circle.

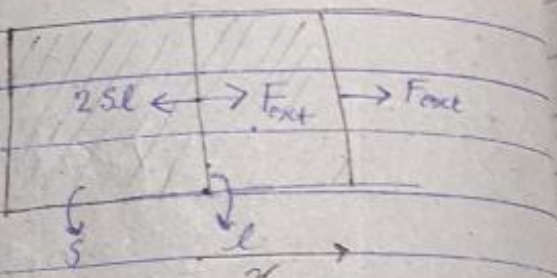
$\therefore$ remaining area will be minimum.

• Surface Energy

moving against some force



molecules at surface will have extra energy known as surface energy.



$F_{ext} = 2Sl$

work done by $F_{ext} = 2Slx$

$2Slx = \Delta U + \Delta K \to 0$

$\Delta U = 2Slx = S(2lx)$

$\Delta U = S\Delta A$

U = surface energy $= SA$

(Q) A water droplet of radius R and surface tension S is divided into 1000 identical droplets find:

(i) Initial surface energy of system.
(ii) Final " " "
(iii) change in " " "

(i) $U_i = SA = S4\pi R^2$

(ii) $\frac{4}{3}\pi R^3 = 100 \frac{4}{3}\pi r^3, \quad r = \frac{R}{10}$

$\therefore U_f = S \times 4\pi \frac{R^2}{100} \times 1000 = 40\pi R^2 S$

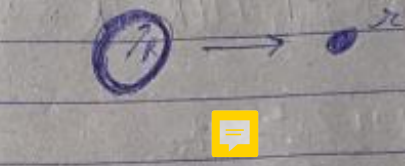
(iii) $\Delta U = U_f - U_i = 36\pi R^2 S$

(iv) If specific heat of liq. = s and density of liq. = f find Δ change in temp when it breaks into 1000 droplets.

$\Delta U = 36\pi R^2 S = \frac{8}{3}\pi R^3 \Delta T s$

$\Delta T = \frac{27S}{8sR}$

(8) A soap bubble of radius R and thickness t ($t << R$) and surface tension S is pricked and converts into droplets. Find :



① initial surface energy.
② final surface energy.
③ change in surface energy.

① $V_i = SA = S 8\pi R^2$

② $V_i = V_f \Rightarrow 4\pi R^2 t = \frac{4}{3}\pi r^3 \Rightarrow r = (3R^2 t)^{1/3}$

$V_f = SA = S 4\pi r^2 = S 4\pi (3R^2 t)^{2/3}$

③ $\Delta U = U_f - U_i = 4\pi S (3R^2 t)^{2/3} - 8\pi S R^2$
$= 4\pi S \{(3R^2 t)^{2/3} - 2R^2\}$

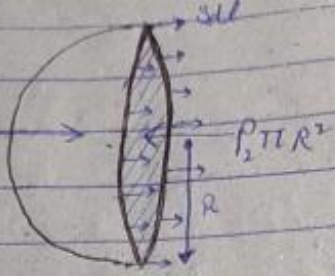
(9) If density of liquid is ρ and specific heat is s Find change in temperature.

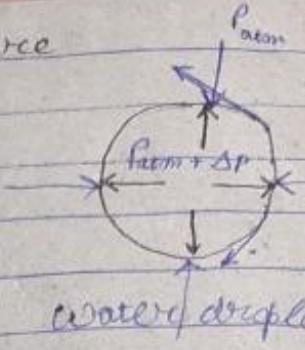
$= ms\Delta T = \rho 4\pi R^2 t s \Delta T$

• Excess Pressure / Gauge Pressure

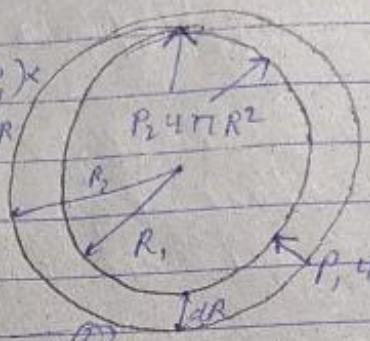
$P_2 \pi R^2 = P_1 \pi R^2 + 2\pi R S$

$$\boxed{P_2 - P_1 = \frac{2S}{R}}$$





$P_2 4\pi R^2 dR - P_1 4\pi R^2 dR = W = (P_2 - P_1) \times 4\pi R^2 dR$ — (1)



$\Delta U = S 4\pi (R^2 + dR^2$

$\Delta U = 4\pi S.(R+dR)^2 - 4\pi R^2 S$

$= 4\pi S\ (R^2 + (dR)^2 + 2R\,dR - R^2)$ — (2)

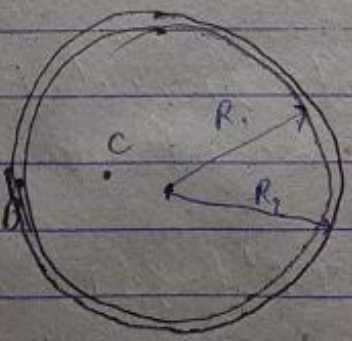
$(P_2 - P_1) 4\pi R^2 dR = 4\pi S\ (2R\,dR) = \Delta P$

$$P_2 - P_1 = \frac{2S}{R}$$

• Excess Pressure in soap bubble

$P_A = P_{atm}$

$P_B = P_{atm} + \frac{2S}{R_2}$



$P_C = P_B + \frac{2S}{R_1}$

$= P_{atm} + \frac{2S}{R_2} + \frac{2S}{R_1}$

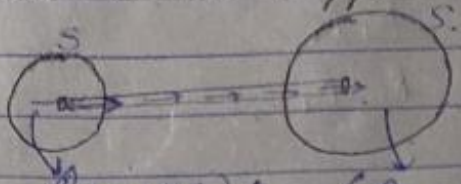
$\therefore P_C = P_A + \frac{2S}{R_1} + \frac{2S}{R_2}$

$$P_C = P_{atm} + \frac{2S}{R_1} + \frac{2S}{R_2}$$

Extra pressure $= \frac{2S}{R_1} + \frac{2S}{R_2} = \Delta P$

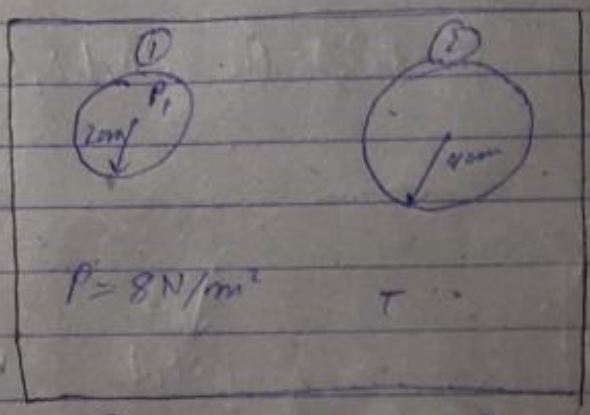
if $R_1 \approx R_2 \Rightarrow \Delta P = \frac{4S}{R}$
(soap bubble)

(8) Two soap bubbles one smaller than other are connected by a straw. What will happen?



consequently small bubble vanishes. And entire air inside soap bubble will go to larger one.

(9) Find pressure inside ① and ②



$S_{bubble} = 0.04\ N/m$

$P_1 = 8 + \frac{4S}{R_1} = 8 + \frac{4 \times 0.04}{2 \times 10^{-2}}$

$P_1 = 16\ N/m^2$

$P_2 = 8 + \frac{4S}{R_2} = 8 + \frac{4 \times 0.04}{0.04}$

$= 12\ N/m^2$

Find ratio of no. of moles of gas inside the bubble.

$n = \frac{PV}{RT} \therefore \frac{P_1 V_1 / RT}{P_2 V_2 / RT} = \frac{P_1 V_1}{P_2 V_2} = \frac{16 \times \frac{4}{3}\pi 2^3}{12 \times \frac{4}{3} \times \pi 4^3}$

$\frac{n_1}{n_2} = \frac{16 \times 8}{12 \times 64} = \frac{1}{6}$

(8) Two soap bubbles of radius R_1 and R_2 touch each other. Find R_c



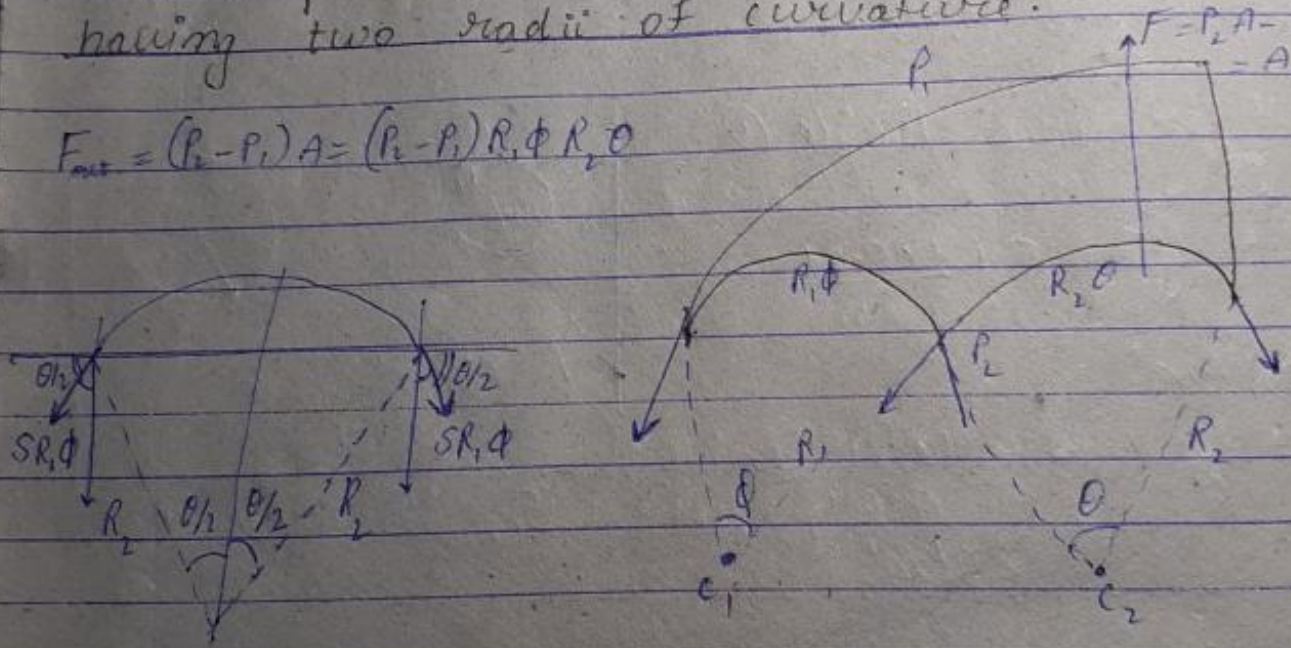
$$P_{atm} + \frac{4S}{R_1} - \frac{4S}{R_c} = P_{atm} + \frac{4S}{R_2}$$

$$\frac{1}{R_c} = \frac{1}{R_1} - \frac{1}{R_2}$$

$$R_c = \frac{R_1 R_2}{R_2 - R_1}$$

- Excess pressure in a curved surface having two radii of curvature.

$$F_{out} = (P_2 - P_1)A = (P_2 - P_1) R_1\phi R_2\theta$$



$$SR_1\phi \sin\frac{\theta}{2} \times 2 = F \Rightarrow \sin\frac{\theta}{2} \text{ and } \sin\frac{\phi}{2} \approx \frac{\theta}{2}, \frac{\phi}{2}$$

$$+ SR_2\theta \sin\frac{\phi}{2} \times 2$$

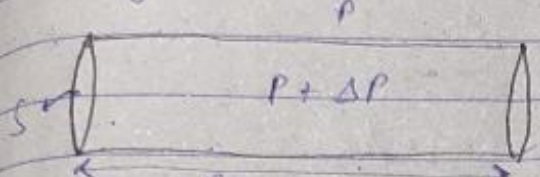
$$\therefore F = (R_1 + R_2)\theta\phi S \; ; \; S(R_1 + R_2)\theta\phi = (P_2 - P_1) R_1\phi R_2\theta$$

$$P_2 - P_1 = \frac{(R_1 + R_2)S}{R_1 R_2} = \boxed{S\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \Delta P}$$

Special cases

① If $R_1 = R_2 = R$ (say) $\Rightarrow \Delta P = \frac{2S}{R}$

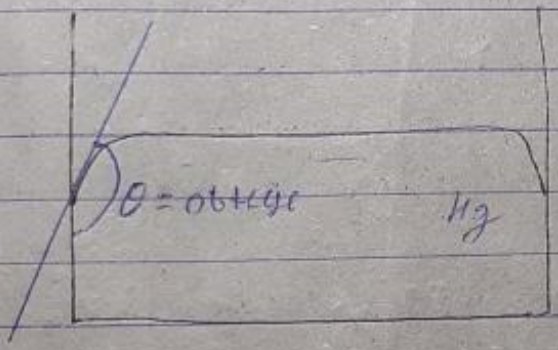
② Cylindrical surface

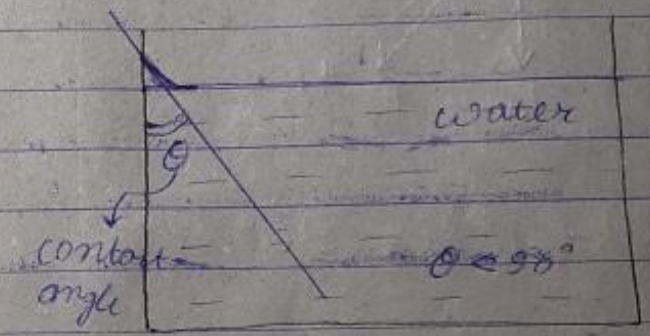


$$\Delta P = S\left(\frac{1}{R} + \frac{1}{\infty}\right)$$

$$\Delta P = \frac{S}{R}$$

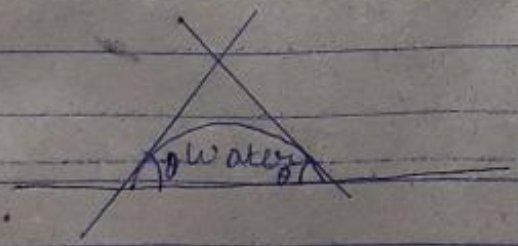
• Contact Angle.

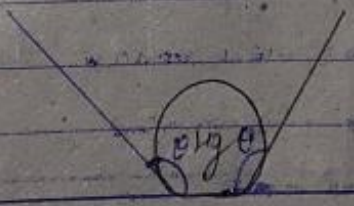




In this case liq. wet the solid.

liquid avoids solid and doesn't wet solid.

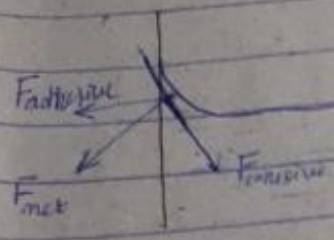


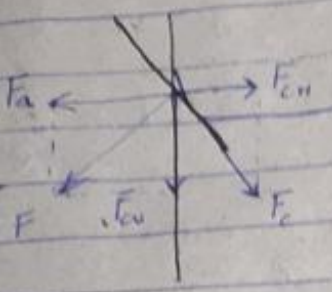


contact angle depends on intermolecular forces. i.e. b/w liquid-liquid molecules and liquid-glass molecules.

when $\theta < 90^\circ$

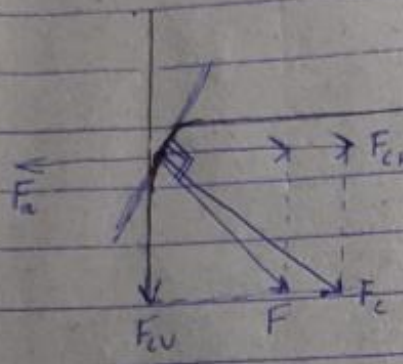
Fadhesive > Horizontal component of cohesive force



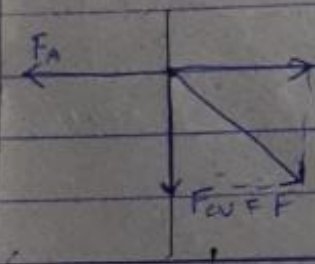


when $\theta > 90^\circ$

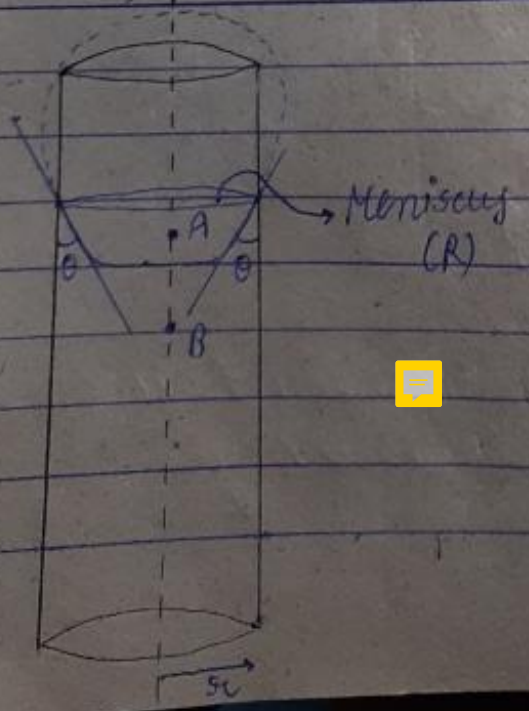
horizontal component of cohesive force must be greater than adhesive force.



when $\theta = 90^\circ$

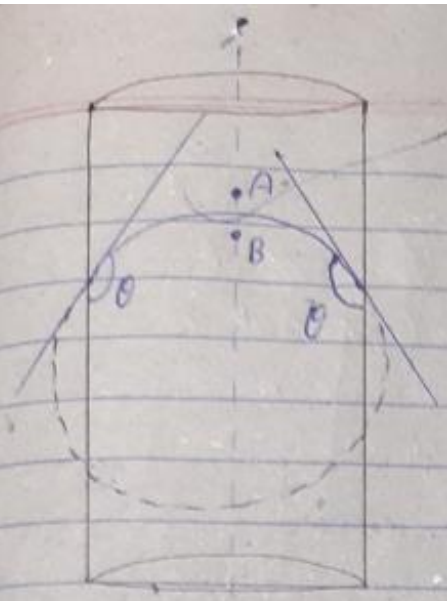


horizontal component of cohesive force is equal to adhesive force.



$$P_B = P_A - \frac{2S}{R}$$

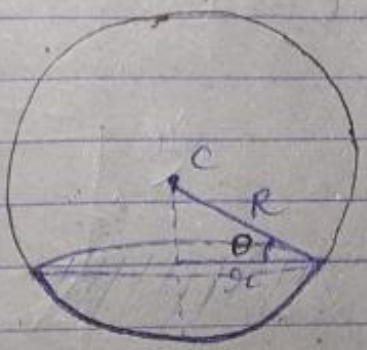
R is radius of curvature of meniscus.

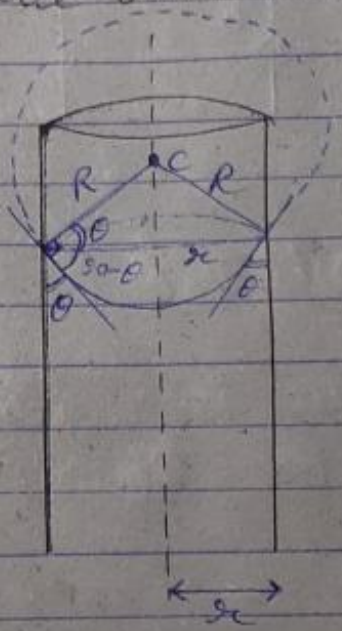


$$P_B = P_A + \frac{2S}{R} = P_0 + \frac{2S}{R}$$

→ Relation between radius of tube and radius of curvature of meniscus (R)

$$\boxed{\cos\theta = \frac{r}{R}}$$

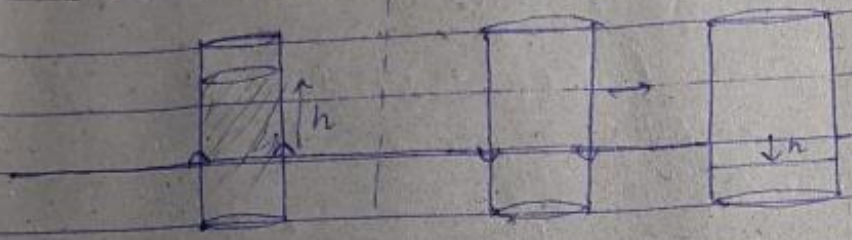




• Capillary Action / Capillarity

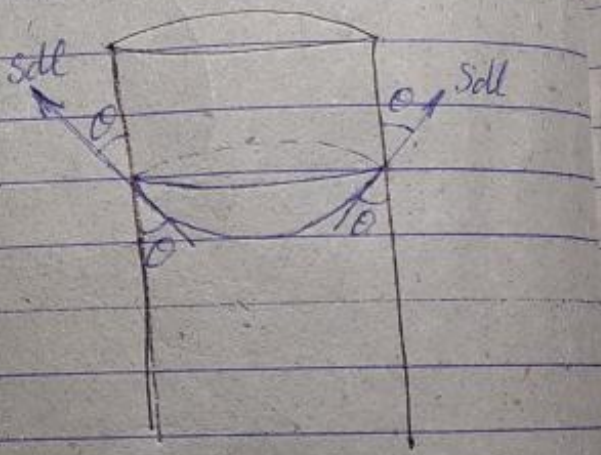
Capillary tube is a tube open from both the ends. (Generally it is cylindrical and is very thin (r is very small). In general made of glass.

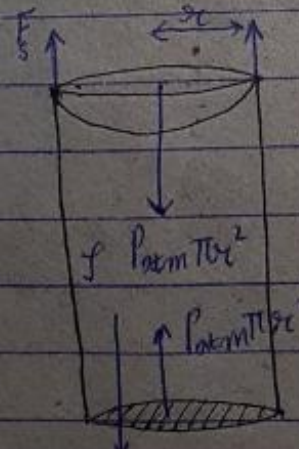
$\theta < 90°$ $\quad$ $\theta > 90°$



$F_s = \int S dl \cos\theta$

This force is responsible for lifting the liquid.

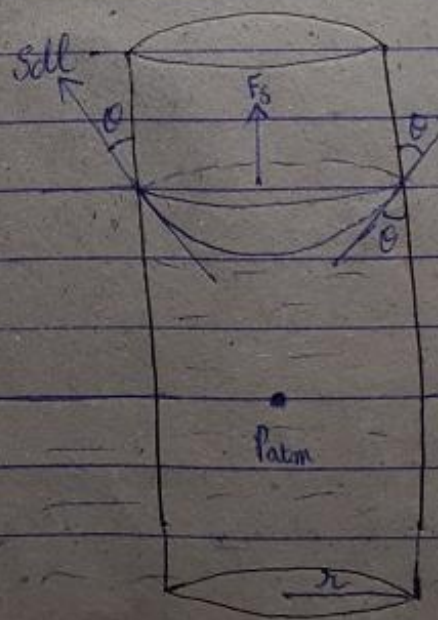




$2\pi r S\cos\theta = f\pi r^2 hg$

$2\pi r S\cos\theta = f\pi r^2 hg$

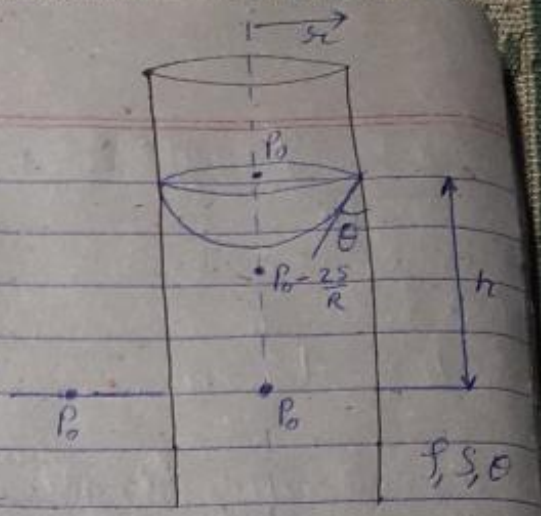
$$h = \frac{2S\cos\theta}{f r g}$$



f, S, θ

$P_0 = P_0 - \frac{2S}{R} + \rho g h$

$h = \frac{2S}{\rho g R} = \frac{2S\cos\theta}{\rho g r}$



(a) $\rho = 10^4\ kg/m^3$, $r = 0.1\ mm$, $S = 0.05\ N/m^2$, $\theta = 143°$

$h = \frac{2 \times 0.05 \times -\frac{4}{5}}{10^4 \times 10 \times 0.1 \times 10^{-3}}$

$\cos 143 = \cos(90+53) = -\sin 53$

$h = -2 \times 4 \times 0.01\ m = -8\ mm$

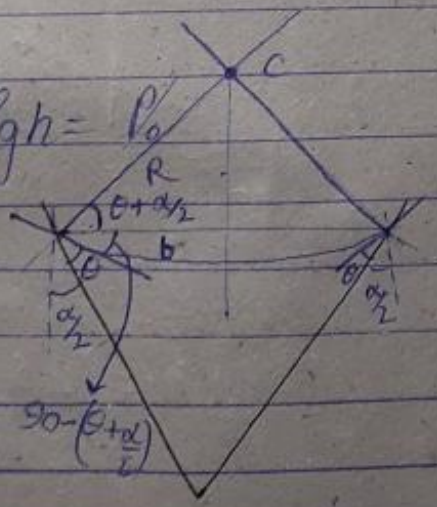
(b) $\rho = 10^3\ kg/m^3$, $S = 0.075\ N/m$, $\theta = 0°$, $h = 15\ cm$.

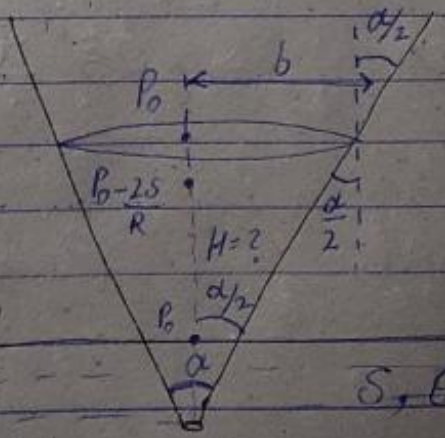
$r = \frac{2S\cos\theta}{\rho h g} = \frac{2 \times 0.075 \times 1}{10^3 \times 0.15 \times 10} = 10^{-4}\ m$

$= 0.1\ mm$

(c) $P_0 - \frac{2S}{R} + \rho g h = P_0$

$h = \frac{2S}{\rho g R}$

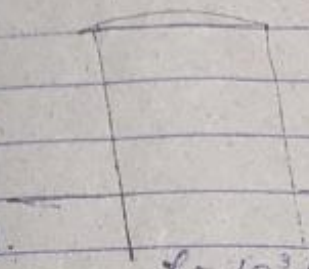




$\cos\left(\theta + \frac{\alpha}{2}\right) = \frac{b}{R} \Rightarrow R = \frac{b}{\cos\left(\theta + \frac{\alpha}{2}\right)} \Rightarrow h = \frac{2S}{\rho g \frac{b}{\cos\left(\theta + \frac{\alpha}{2}\right)}}$

What if capillary tube is of insufficient length

(Q) Find h
$h = 16cm$



$\rho = 10^3 kg/m^3$
$\theta = 0°, S = 0.08 N/m$
$r = 0.1mm$

If 8cm length is taken then liquid then or less than 16cm.
① liquid will rise upto top and stops.

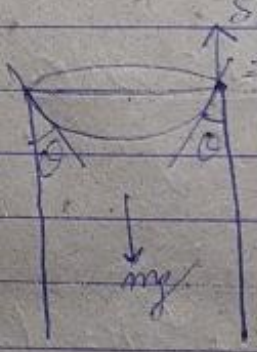
② wont overflow as contact angle will increase. $\Rightarrow \cos\theta \downarrow \Rightarrow F_s \downarrow \Rightarrow F_s = mg$

(Q) In previous ques. if length of capillary tube above water surface is only 8cm, what will happen?

$$\cos\theta = \frac{\rho r g h}{2S}$$

$$\cos\theta = \frac{10^3 \times 10^{-4} \times 10 \times 0.08}{2 \times 0.08}$$

$$\cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$



Actual
$$h = \frac{2S\cos\theta}{\rho r g}$$

Insuff. length of tube
$$h' = \frac{2S\cos\theta'}{\rho r g} < h$$

$$\frac{h'}{h} = \frac{\cos\theta'}{\cos\theta}$$